Lyracist Pro Suite

Complete User Manual & Reference Guide

Lyracist Pro, LyracistDbEditor, and KSRotation Displays



Version 26.7 (July 2026 Release)

PAROLE Software Co.

1. Introduction to the Lyracist Pro Suite

Welcome to Lyracist Pro, the ultimate digital hosting and rotation management software suite tailored specifically for professional Karaoke Jockeys (KJs), DJ hosts, and entertainment venues. The suite is composed of three interconnected modules designed to coordinate lyrics playback, performer request flows, rotation scheduling, and public screen displays:

- Lyracist (Main App): The central desktop workspace where the host controls audio adjustments, video decoding, active singer rotations, catalog scanning, settings, and the mobile requests inbox.
- LyracistDbEditor: A standalone database utility used to browse, clean, import, and configure SQLite database files offline, ensuring no write conflicts or performance hits during live shows.
- KSRotation (Display Renderer): The projection renderer window that drives secondary and tertiary display monitors, rendering real-time lyrics overlay screens and custom themed rotation billboards.
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- [IMAGE PLACEHOLDER: System Topology Setup]
- Graphic Details: A block diagram showing the main Lyracist hosting application running on a laptop, connected via HDMI to Monitor 2 (performer lyrics) and Monitor 3 (public rotation billboard), while local performer mobile phones connect via Wi-Fi to the local SignalR web server portal.

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2. Lyracist (Main Host Application)

Lyracist runs as a high-performance desktop application built on modern WPF (.NET 10.0 Windows) architecture. It functions as the host's control dashboard, integrating music players, singer queue management, and live vocal DSP chains into a single interface.

2.1. The Main Dashboard & Playback Controls

The Live Deck sits at the top of the interface and controls playback of the current performer's track. It is driven by a high-performance LibVLC decoding engine to ensure maximum compatibility with video and CDG zip backdrops.

- Play / Pause Button: Starts or suspends the loaded song track. Can be triggered globally via the mapped Play/Pause hotkey.
- Stop Button: Halts playback completely, resets the seek bar, and triggers ambient background or fill-in music automatically.
- Seek Slider: Allows the host to click and drag to jump to any part of the active track.
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- Drive Status Indicator: A glowing LED and text indicator on the main window title line showing whether library folders/drives are currently connected or missing. A green light (Drives Online) indicates all library folders are reachable, yellow indicates some folders are offline or no folders are configured, and red (Drives Offline) alerts you that the external storage drive is disconnected.
- Volume Slider: Adjusts the master volume output (0% to 100%) in real time.
- Speed (Tempo) Slider: Alters playback speed from 0.5x to 2.0x without shifting the audio pitch.
- Pitch Transpose Slider: Adjusts the key of the backing track in semitone steps from -6 to +6.
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- [IMAGE PLACEHOLDER: Main Dashboard Controls]
- Graphic Details: A high-fidelity screenshot of the Lyracist main window, indicating the Play/Pause, Stop buttons, Seek slider, and deck speed/pitch controls highlighted with callout boxes.

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2.2. Performer Queue & Rotation Management

The singer queue is built around a round-robin rotation algorithm to ensure that every singer gets a fair turn. The list displays active singers, their requested songs, transposition overrides, and status flags.

- Adding Singers: Enter the singer's name and select a song from the library, online Party Tyme stream, or custom external link (YouTube/Spotify).
- Duet Partner Option: An optional second singer slot. Enabling a duet prompts the system to configure dual audio profiles.
- Done Singer Command: Triggered when a singer finishes performing. It logs the performance to the singer's history, advances them to their next queued song, and moves them to the bottom of the list.
- Reordering: Drag and drop queue items to manually override the rotation order, or use the Move Up / Move Down buttons.

2.3. Dual-Singer Duet Audio & Vocal DSP Settings

When a duet is detected (a song is loaded with both a main singer and a duet partner), Lyracist Pro automatically instantiates individual DSP channels for both performers. The Vocal Settings dialog dynamically presents two configuration tabs: 'Main Performer' and 'Duet Partner'. Each profile allows independent control over:

- Vocal Gain (dB): Compels matching mic input volume amplification.
- Transposition Preference: Individual key transpose offset if singers require different key adjustments.
- 3-Band Parametric EQ: Independent sliders to boost or cut Bass, Midrange, and Treble frequencies (+/- 10dB) to suit the vocal characteristics of each singer.
- Dynamic Compressor Threshold: Applies automatic volume leveling to manage peaks and whispers.

- Peak Limiter Threshold (dB): Caps maximum volume to protect the sound system against clipping and microphone drops.
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- [IMAGE PLACEHOLDER: Duet Vocal Settings Window]
- Graphic Details: A screenshot of the Vocal Settings popup showing two tabs, 'Main Performer' and 'Duet Partner', with independent sliders for 3-band EQ, compressor, and peak limiter.

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2.4. Venue Config & Customizable Transition Countdown

The Settings tab contains options to customize venue-specific branding and timing parameters:

- Venue Selector: Choose the current venue. Each venue saves its own local preferences and graphics folder path.
- Venue Slideshow Graphics: Add multiple graphic files (.jpg, .png) for each venue. When the rotation is idle, the public screen cycles through these graphics in a background slideshow loop.
- Auto-Advance Transition: Toggle the automatic transition countdown. When a singer completes a song, the system begins a visual countdown (customizable from 5s to 30s) before advancing to the next singer.
- Done Timer Overlay: Public display displays a ticking countdown bar. The KJ can click 'Advance Now' to bypass, or 'Cancel Countdown' to keep the screen paused.

2.5. Keyboard Shortcut Mapper (Learn Mode)

The 'Keyboard Hotkeys' tab allows the host to bind keyboard keys to main deck commands. To map a command, select the text field next to the action and press the desired key on the keyboard. Supported actions include Play/Pause, Stop, Done/Next Singer, Toggle Crawl Banner, Show Lyrics, and Show Rotation. The mapping system contains a built-in safety filter: hotkeys are automatically bypassed when the cursor is focused inside search fields or text inputs to prevent accidental triggers while typing names.

[IMAGE PLACEHOLDER: Keyboard Shortcut Learn Panel]

Graphic Details: A screenshot of the Settings - Keyboard Hotkeys tab showing input mapping boxes for Play/Pause, Stop, Done Singer, and Show Lyrics, with mapped keys like Space, Escape, and Return.

2.6. Product Registration Activation

Lyracist Pro features offline cryptographic license verification. In the Settings - Registration tab, the host enters their First Name, Last Name, Stage Name, Email Address, and Serial Key. The key is matched

against a SHA-256 + Custom Base-32 keygen algorithm. The validation state renders visual cards: a green checkmark (CheckmarkCircle24) indicates Lyracist Pro is fully activated; a red cross (DismissCircle24) represents validation failure; and a grey key represents awaiting inputs.

2.7. Integrated Mobile Request Portal & Wi-Fi Server

Lyracist Pro runs a private local web server via SignalR. Performers connected to the venue's local Wi-Fi can scan the QR code displayed on the screen to open the mobile web portal. Without needing internet access, performers can browse the host's local song catalog, view the current singer rotation queue, submit song requests directly to the host's Inbox, and spin the Scaryoke Wheel of Fortune to get a randomly assigned category and song.

Performer requests are split into Karaoke requests (singer performing) and background Music requests. A separate 'Search Music' tab is provided on the mobile portal for browsing the background music collection. In addition, performers can tap reaction buttons (👏 , 🔥 , ❤️ , 🙌 , 🎉 , 👑) to send live, animated, color emoji reactions that drift and fade across both projection screens in real time.

2.8. LYRX Parity Visual & Audio Enhancements

We integrated two major performance features from LYRX Karaoke: AI Vocal Suppression ('Kill Vocal') and the Audio-Responsive Visualizer. The AI Vocal Suppression uses native LibVLC EQ mid-band cancellation (500Hz to 4kHz) to suppress lead vocal tracks by -20dB, transforming normal audio/video files into instant backing tracks. The Audio-Responsive Visualizer renders neon-gradient spectrum bars that react in real time at the bottom of the public lyrics window when playing, or slow ocean waves when idle.

[IMAGE PLACEHOLDER: Live Deck with Kill Vocal Toggle]

Graphic Details: A screenshot of the deck playback panel highlighting the 'Kill Vocal (AI)' toggle switch situated next to the 'Reset Audio' button.

2.9. Audio Outputs & Hardware Mixer Mode

Lyracist Pro supports routing audio streams to separate physical output channels on external hardware mixers (e.g., Pyle PMXU88BT).

- Karaoke Audio Output: The output channel dedicated to playing the singer's backing track, configured in settings.

- Background Music (BGM) Output: The output channel dedicated to ambient fill-in, opening, and end-of-rotation playlists, keeping background music independent of the performer's microphone channel.

- Hardware Mixer Mode: Bypasses all software DSP volume, EQ, compressor, and limiter parameters, sending a clean, flat 100% volume signal directly to the external physical mixer to avoid muddy sound from double-processing.

2.10. Performer XP, Levels & Progression

The suite features a built-in gamification system for performer engagement. Experience points (XP) are calculated as: XP = Total Songs Sung * 100 + Average Performance Score. Singers automatically unlock performance titles (e.g., Shower Singer, Pub Regular, Vocal Powerhouse, Karaoke Legend) and achievement badges (Debut, Legend, Rising Star, Crowd Pleaser, High Scorer) visible on the rotation billboard and tablet portal leaderboards.

3. LyracistDbEditor (Offline Catalog Manager)

LyracistDbEditor is a utility designed to manage the SQLite databases offline, avoiding locks or interference while the main show is running.

- Library Importer: Import directories of MP3, WAV, MP4, and zipped MP3+G files. It automatically extracts song titles and artist names from file names based on customizable templates (e.g. 'Artist - Title' or 'Title - Artist').
- Singer Database: View and edit all performer records. You can override completed song counts, last-sang timestamps, and clear ratings.
- Slideshow Setup: Map and verify venue graphic slideshow paths to ensure they match local directories.
- Database Purging & Backups: Create compressed database backups before major show events, or purge old history entries.
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- [IMAGE PLACEHOLDER: LyracistDbEditor Catalog Panel]
- Graphic Details: A screenshot of the database editor showing the song library grid, search filters, and import folder directory settings.

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4. KSRotation (Public Display Projection)

KSRotation controls the secondary and tertiary video outputs. It renders lyrics and rotates the performer billboard display modes.

- Normal Queue Mode: Renders a modern, clean dark-themed list of upcoming performers.
- Star Wars Crawl: Renders the performer queue as a 3D perspective scrolling text crawl, scrolling into deep space with customizable intro banners ('A long time ago, at a karaoke bar far away...').
- Vegas Marquee Mode: Showcases the current and next performer names inside an animated, glowing neon lightbulb marquee grid.
- Vinyl Turntable Mode: Renders a spinning record sleeve graphics template with track details.
- Transition Auto-Advance Countdown: Displays a progress bar and remaining seconds ticker on the public screen when a song finishes, letting performers know when the next singer is about to launch.
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- [IMAGE PLACEHOLDER: Star Wars Crawl Display Mode]
- Graphic Details: A rendering of the public projection screen showing the performer queue scrolling as a 3D Star Wars text crawl into space, complete with background stars.

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Section: Database Manager / Catalog Book Exporter

[Update Details]

Operators can now export the entire song database (Karaoke or Music) into Word (.docx) or PDF format directly from the Lyracist Database Manager.

1. Exporter Panel:

- Located in the right pane of the Lyracist Database Manager under "Catalog Book Generator".

- Provides two dropdown selectors:

- Catalog: Choose between the "Karaoke Catalog" (tracks marked as karaoke) and the "Music Catalog" (standard audio tracks).

- Format: Choose between "Word Document (.docx)" and "PDF Document (.pdf)".

2. Generated Fields & Formatting:

- The generated books are structured as clean, professionally styled tables showing:

- Artist: Alphabetically ordered performer names.

- Title: Song titles.

- Length: Formatted track durations (MM:SS).

- Genre: The embedded audio genre (automatically populated during metadata scans).

3. Pagination & Output:

- The document is fully paginated, displaying "Page X" or "Page X of Y" in the bottom footer.

- For Word (.docx) books, a repeating header row is defined so column titles automatically reappear on the top of each page.

- When generation completes, Lyracist automatically launches the default system viewer to open and print the document, which is saved in the "Reports" directory.

Section: Display Projection Window / DJ Banner Screen & Selection

[Update Details]

Operators can now configure and display custom DJ branding, logos, and promotional banners on a designated projection screen.

1. DJ Banner Screen Target:

- Available in both Lyracist (Settings page & Karaoke page) and KSRotation (Settings tab).

- Allows operators to choose which connected monitor will project the borderless, full-screen DJ banner.

- When enabled/opened, it launches the screen on the target monitor and keeps it topmost.

2. Upload Banners:

- Operators can upload multiple banner images (PNG, JPG, JPEG, GIF, BMP) from their system via the 'Upload' button.

- Files are copied into a local DJBanners settings subdirectory automatically.

3. Select Active Banner:

- If more than one banner is uploaded, operators can select which banner is active and displayed from a dropdown list of available images.

- Selecting a different banner updates the projected screen instantly without reloading the window.

Section: Display Projection Window / Casting Support

[Update Details]

Operators can now select different casting targets for the Singer Rotation Billboard screen in both Lyracist and KSRotation applications, in addition to standard physical monitors.

1. Casting Target Dropdown:

- Located in the "Settings" tab under "Displays & Projection" (or "Display & Projection" in KSRotation).

- Dropdown options include Monitor, Miracast, Chromecast, Browser Cast, AirPlay, and Wireless HDMI.

- When a casting target is selected, the application (Lyracist or KSRotation) initializes the appropriate casting sender/server in the background.

2. Off-Screen Rendering:

- Selecting a casting target (such as Chromecast or Browser Cast) creates and renders the Rotation Window / Singer Display Window in an off-screen buffer (invisible to the host desktop) to capture frames dynamically.

- This ensures the KJ laptop screen is not cluttered while casting frames.

3. Browser Cast Server:

- The self-hosted Browser Cast option spins up a local web server (http://localhost:8080/rotation/) to allow any browser on the local network to view the singer rotation billboard in real-time.

Section: Database Manager / Catalog Book Exporter & Failed Artist List

[Update Details]

Updates have been made to the Catalog Book Exporter formats and the Slow Metadata Extractor.

1. Supported Formats and Sorting:

- The Catalog Book Exporter now supports exporting the catalog as a plain text (.txt) file.

- The format selection dropdown has been sorted in the following order: PDF Document (.pdf), Text File (.txt), and Word Document (.docx).

2. Metadata Scan Failed Artist Export:

- When running a Slow Metadata Scan for songs with unknown artists, any files that were unable to be resolved and updated with a valid artist during the scan are tracked.

- Upon completion or cancellation of the scan, a plain text file listing all unresolved file paths is automatically exported to the "Reports" directory, named "Failed_Artist_Updates_[Timestamp].txt".

Section: Integration Settings / Party Tyme Removal

[Update Details]

The Party Tyme karaoke streaming integration has been removed from the application.

1. Removal of Streaming Tab:

- The "Party Tyme Online" tab has been removed from the main Karaoke dashboard view.

- The "Party Tyme (Streaming)" tab has been removed from the Special Occasions Music search dialog.

2. Settings Panel Clean-up:

- The subscription fields (Client ID and Client Secret) for Party Tyme have been removed from the Settings panel under APIs & Logins.

3. Queue & Badge Updates:

- The "Party Tyme" filter checkbox and visual badge triggers have been removed from the singer rotation queue list.

Section: Performer Profiles, Song Tagging & Singer Avatars

[Update Details]

A comprehensive performer profile system and song tagging support have been integrated into Lyracist.

1. Performer Profiles & Login:

- Patrons accessing the Patron Request Portal can log in or register using their Performer Name and a 4-digit PIN.

- Profile settings include email registration, custom vocal range specification (e.g. Tenor, Soprano), and custom title assignment (e.g. The Screamer).

2. Singer Avatars & Custom Uploads:

- Performers can customize their profile avatar using a dropdown selector.

- Supports default silhouette, Gravatar logo generation (via email MD5 hash), and direct mobile photo/selfie uploads (capped at 2MB).

- Avatars are displayed as circular icons in the rotation board on both the mobile Patron Portal and host WPF Singer Queue ListBox.

3. Database Tagging:

- Songs can now be tagged with comma-separated categories to organize, filter, and challenge performers.

// Edited on Aug 19, 2026 @ 11:24:00 -> Added Help & About tabs in LyracistTrivia and timer override documentation

Rotation Management & Round Cycle Tracking (`Lyracist`, `KSRotation`, & `KSRotation.Maui`) - (August 2026)

Section: Performance & Rotation Operations / Singer Queue Management

[System Details]

1. Float Current Singer to Top of Rotation:

- **User Interface Controls**:

  * `Lyracist`: Checkbox on the **Rotation** page queue toolbar, **Karaoke** page queue bar, and persistent toggle under **Settings -> Monitors & Screen Assignments**.

  * `KSRotation` (WPF): Checkbox on the main rotation toolbar with automatic settings persistence.

  * `KSRotation.Maui`: Checkbox in the Active Rotation Queue header bar.

- **Queue Behavior**:

  * When enabled, the performer currently at the microphone automatically floats to the very top (index 0) of the list.

  * Upon song completion (`Done` / `✓`), the finished performer is moved to the back of the active queue and the next active performer automatically floats to the top (index 0).

  * Eliminates the need for the DJ/KJ to scroll down long lists during active shows.

2. 1st Singer in Rotation (Round Start Anchor Flag):

- **Visual Indicator**:

  * A red `🚩 1ST` badge is displayed next to the singer designated as the first performer of the rotation round, and that singer's row is also highlighted with a red background (alongside the existing yellow "now performing" and blue "up next" row highlights), so the 1st singer stands out at a glance.

  * When the 1st singer returns to index 0 / current status, the DJ knows a complete rotation cycle has been completed.

- **Manual Re-anchoring & Clearing**:

  * Right-clicking any singer and selecting `"Set as 1st Singer (Round Start)"` or clicking the `🚩` button reassigns the anchor to that singer, clearing it from previous holders.

  * Clicking the same action again on the singer who already holds the flag (the button turns red / the menu item changes to `"Clear 1st Singer Badge"`) undoes the mistake — it hands the flag to the next active singer in rotation order rather than leaving nobody flagged, so an accidental click never needs a specific replacement singer picked.

  * When new singers are added, the system automatically ensures exactly one active singer holds the start flag.

- **Cross-App Sync**:

  * The `isRotationStart` flag is synchronized via the REST/WebSocket `/api/rotation` feed between `KSRotation` and `Lyracist`.

Lyracist Live Trivia (`Lyracist`, `KSRotation`, & `Lyracist.Trivia`) - Integrated Game Master & Intermission System (August 2026)

Section: Entertainment Modules / Lyracist.Trivia & Pub Trivia Game Master

[System Details]

1. Multi-Monitor Big Screen Projection & Auto-Casting in `KSRotation`:

- **" 📡 Open Big Screen" Button**:

  * Directly on the **` 🎯 Trivia Game Control`** tab in `KSRotation`.

  * Toggles and focuses the full-screen 16:9 venue projection window (`TriviaDisplayWindow`) on the selected target monitor.

- **Automatic Projection on Game Start**:

  * Clicking **"▶ Start Game"** in `KSRotation` automatically opens and positions the trivia projection window onto the designated secondary monitor/TV so patrons can immediately see questions, timer countdowns, and options.

- **Dynamic Repositioning**:

  * If the target monitor is switched in display settings, `TriviaDisplayWindow` dynamically relocates to the newly selected monitor.

2. Hybrid Game Engine Architecture & Integrated Host Decks:

- **`Lyracist` Application Integration**:

    * Top-level navigation items for **`Trivia`** (interactive host deck) and **`Trivia Settings`** (3-column no-scroll dashboard for timers, points, bonuses, and Wi-Fi credentials).

  - **`KSRotation` Application Integration**:

    * Dedicated main window tabs for **`Trivia`** and **`Trivia Settings`**.

  - **Deduplicated Venue & Game Master / DJ Settings**:

    * Trivia settings in both apps pull directly from existing Venue (`AppSettings.SelectedVenue` / `MainViewModel.VenueName`) and DJ / Host (`AppSettings.DjName` / `MainViewModel.DjName`) properties.

    * Duplicate textboxes were removed and replaced with clean synchronized information badges.

  - **3-Column No-Scroll Trivia Settings Dashboard**:

    * Divided into 3 horizontal columns:

      - Column 0: Question Timers & Flow, Scoring & Bonus Multipliers (Base Points, Wrong Answer Deduction Points [Default: 0], Speed Bonus, Streak Multipliers)

      - Column 1: Active Venue & Host Information Badge, Intermission & Pre-Game Countdown

      - Column 2: Local Network & Wi-Fi Access, Save / Reset Action Cards

    * Eliminates vertical scrolling on standard desktop resolutions.

  - **Automated Game Pause & Resume Synchronization**:

    * Automatically pauses the trivia round when DJ Banners, Special Events, or Rotation screens take over the projection display.

    * Displays `"Game Paused: {Reason}"` on the host console and `#pause-banner` on mobile buzzers, safely freezing timers and blocking submissions until the overlay is closed.

  - Run as a standalone application (`Lyracist.Trivia`) for dedicated venue Trivia Nights, or trigger rapid fill-in / intermission rounds directly during karaoke shows.

  - Shared core library `Lyracist.Trivia.Core` handles SQLite persistence, real-time scoring, speed bonuses, streak multipliers, and player socket communication.

  - **Venue & Game Master Branding Configuration**:

* Configure Venue Name (e.g., "The Neon Lounge") and Game Master / Host Name (e.g., "Trivia Master") on the ** ⚙️ Settings & Display** tab.

* Updates dynamically across all projection screens, scoreboard banners, and mobile web buzzers in real time.

- **Multi-Monitor Display Resolution & Placement**:

* Select any connected monitor (Monitor 1, Monitor 2, HDMI TV, or projector) from the Game Master console.

* Uses DPI-aware screen bounds positioning without forcing to Monitor 0.

- **Projection Screen Escape & Close Controls (`TriviaDisplayWindow`)**:

* **Escape Key (`Esc`)**: Pressing Escape immediately closes and exits the projection display window.

* **Floating Close Button (`✕`)**: Sleek translucent circular close button in the top-right corner, illuminating red on hover for instant 1-click closure.

* **Right-Click Context Menu**: Right-clicking anywhere on the screen displays quick options to close (`✕ Close Projection (Esc)`) or toggle fullscreen (`🗖 Toggle Fullscreen (F11)`).

* **Double-Click & Drag**: Double-clicking the display toggles between windowed and fullscreen maximized modes, with click-and-drag window repositioning.

- **Unified 16:9 Pre-Game Lobby & Category Showcase Screen (`TriviaDisplayWindow`)**:

* Combines all pre-game elements into a clean, modern split layout:

1. **Left Hero Column (70% width)**: Full-featured 16:9 Category Announcement Banner (`TriviaData/Banners/{pack}.png`) with ambient glow, tonight's category theme title, and topic subtitle. Cross-fades dynamically when the host selects a different pack in the console dropdown.

2. **Right Onboarding Stack (30% width)**:

- ** ⏱️ Game Start Countdown Clock**: Large digital timer with pulsing badge (`"⏱️ GAME STARTS IN: mm:ss"`) that counts down to launch.

- ** 📶 1. Connect to Wi-Fi Card**: Dedicated scan-to-connect Wi-Fi QR code (`WifiQrCodeImage`), SSID name, and password.

- ** 📱 2. Join Trivia Game Card**: Dedicated scan-to-join mobile buzzer QR code (`QrCodeImage`), direct URL, and buzzer onboarding instructions.

3. **Bottom Connection & Copyright Bar**: Displays mobile buzzer URL (` 📱 PLAY ON YOUR PHONE: ...`), company copyright information (`© 2026 PAROLE Software - All rights reserved.`), and app branding across all projection screens.

4. **Continuous Scrolling Marquee**: Bottom/top ticker bar displaying venue branding, Game Master name, game rules, and buzzer tips.

* **Single-Click Game Master Setup (` 🎯 Launch Pre-Game Lobby & Countdown`)**:

- A single primary action button in the Game Master console that opens/focuses the big screen, locks the category banner, starts the pre-game countdown, and activates the lobby.

- Accompanied by `"▶ Start Game Now (Skip Countdown)"` for immediate question launch.

- **Top Scrolling Player & Score Marquee**:

* Continuous horizontal marquee ticker running along the top of the venue display screen (`TriviaDisplayWindow`).

* Shows all registered players and teams with their live point totals.

* The leading top scorer is dynamically highlighted with a distinctive gold crown badge (`"👑 1. Player (Team): 3,450 pts"`).

- **3-Tab Game Master Console & Settings Page**:

* ** 🎯 Live Game Master Tab**: Streamlined 3-column live command deck with question pack selection, auto-run switch, live question & 2x2 options visualizer, QR connection code, live scoreboard, and round flow controls.

* ** ⚙️ Settings & Display Tab**: Full configuration page for Intermission delay (configurable minutes), auto-start toggle, display monitor selector, question timers (15s), wrong answer elimination speed (5s), post-reveal buffer (5s), base points, speed bonus toggle, sound effects, server port, venue branding, and persistent "Save All Settings" button.

* ** 👥 Players & Teams Tab**: Roster grid showing player names, team names, total points, streaks, correct answer counts, total answered, and individual player kick/removal controls.

- **Post-Game Intermission Countdown & Automatic Next Game Start**:

  * When a game finishes, if auto-start is enabled, an intermission countdown runs for the configured duration (e.g. 3 minutes).

  * Venue display, GM console, and mobile web app display a live ticking countdown banner (`"🎮 NEXT TRIVIA ROUND STARTS IN: mm:ss"`).

  * Host can click `"▶ Start Now (Skip)"` to bypass the intermission, and once expired, the engine automatically selects the next question pack (or reshuffles) and launches the new round.

- **Connect Instruction Screen & Dedicated Trivia Navigation in `Lyracist` and `KSRotation`**:

  * **`Lyracist`**: Access `"👁️ Preview Connect Screen"` in `SettingsPage` (Column 4: Connect & Request Instructions), with full Trivia controls and rules configuration residing in dedicated **Trivia** and **Trivia Settings** navigation pages.

  * **`KSRotation`**: Preview the Wi-Fi and Request Connect screen directly from `Connect & Request Instructions` under the **Display** settings tab, while Trivia rounds and configuration remain cleanly dedicated to the **Trivia** and **Trivia Settings** tabs without toolbar clutter.

- **Configurable Questions Per Game**: Set the number of questions per game round (5, 10, 15, 20, 25, 50, 100, or custom editable value in the host console) stored in `trivia_settings.json`. Slices the shuffled question pack to the exact number of desired questions.

- **Game Complete Winner Celebration & Team Announcements**:

  * Upon completing all allotted questions, the game transitions to a celebration screen on venue big screens (`TriviaDisplayWindow`) and player mobile devices (`trivia.html`).

  * Announces the winning champion and final score.

  * If the winner is a team, prominently displays `"👑 WINNING TEAM: {TeamName}"` and lists all players on that team (`"Team Players: Alice, Bob, Charlie"`).

  * Mobile phones display the winning team/player, team member list, the player's personal placement and final score, and the complete top 10 leaderboard.

- **Zero-Config LAN Mobile Web Server (`TcpListener`)**: Direct socket HTTP server listening on `0.0.0.0` (all interfaces) with RFC 1918 LAN IP resolution, allowing

smartphones and tablets on local Wi-Fi to load buzzers and buzz in without Windows URL reservation restrictions or admin privileges.

- **Dynamic Question Shuffling & 2,100 Total Questions**: When any category is selected or played, the entire question database (150 curated questions per pack across all 14 categories, totaling 2,100 questions) is dynamically shuffled, ensuring questions appear in a fresh, non-repeating order each time a category is chosen.

- **Target Monitor Selector**: Host console features a monitor selection dropdown (Monitor 1, Monitor 2, TV/HDMI) allowing dynamic window positioning and DPI-aware full-screen projection onto any connected venue monitor.

- **Auto-Running Game Loop**:

  * **15-Second Answering Window**: When a question begins, players have 15 seconds to buzz in on their smartphones or venue screens.

  * **5-Second Sequential Wrong Answer Elimination**: When the 15-second window expires, incorrect answers progressively fade out one every 5 seconds until only the correct answer is left standing and illuminated.

  * **5-Second Post-Reveal Buffer & Progression**: Once only the correct answer is illuminated with full explanation and point calculations, the game waits 5 seconds and then automatically advances to the next question.

2. Centralized Trivia Storage (`TriviaData/`):

- Located directly in the solution root at `TriviaData/`.

- Contains SQLite database `trivia.db`, host settings `trivia_settings.json`, and question pack directory `TriviaData/packs/`.

- **15 Comprehensive 150-Question Category Databases (2,250 Questions Total, No Duplicates)**:

  * **Famous Lines & Sayings From Movies** (`famous_movie_quotes.json`): 150 questions covering AFI Top 100 movie quotes, iconic catchphrases, and unforgettable dialogue across Hollywood history.

  * **Bikers & Motorcycles** (`biker_trivia.json`): 150 questions covering Harley-Davidson engines, Indian heritage, European/Japanese classics, rallies, and chopper mechanical lore.

* **Rock & Roll** (`rock_and_roll.json`): 150 questions covering classic rock, 70s/80s bands, albums, and guitar legends.

* **Country Music** (`country_music.json`): 150 questions covering outlaw country, 90s hits, and Opry legends.

* **Geography** (`geography.json`): 150 questions on continents, oceans, mountains, rivers, and landmarks.

* **State Capitals** (`state_capitals.json`): 150 questions covering all 50 US State Capitals + world capitals.

* **History** (`history.json`): 150 questions on world & US history, ancient civilizations, and major milestones.

* **Complete the Lyric** (`complete_the_lyric.json`): 150 singalong lyric questions across rock, pop, and karaoke favorites.

* **TV Shows** (`tv_shows.json`): 150 questions covering classic sitcoms, 90s hits, drama series, and cartoons.

* **Sports** (`sports.json`): 150 questions covering NFL, MLB, NBA, NHL, soccer, Olympics, and golf.

* **Logos & Brand Slogans** (`logos_and_slogans.json`): 150 questions covering advertising taglines, mascots, and brand history.

* **Music & Karaoke Legends** (`music_legends.json`): 150 questions celebrating iconic vocalists, chart-topping legends, and karaoke hall-of-fame artists.

* **80s & 90s Pop Culture** (`pop_culture_80s_90s.json`): 150 questions on retro toys, video game classics, 80s/90s movies, and nostalgic trends.

* **Blockbuster Movie Soundtracks** (`movie_soundtracks.json`): 150 questions covering blockbuster film scores, Oscar-winning theme songs, and needle-drops.

* **Pub Trivia All-Stars** (`pub_general_knowledge.json`): 150 questions spanning science, literature, food & drink, myths, idioms, and pub favorites.

3. Mobile Player Portal (`trivia.html`):

- Zero-installation mobile web app hosted directly by the embedded web server (default port 8085).

- Patrons scan an on-screen QR code on their smartphones to register their player/team name and buzz in using 4 large, responsive touch cards (A, B, C, D) with haptic feedback.

- Live rank indicators, streak counters, and real-time progressive elimination fading on phone buzzers.

4. 16:9 Venue Display Projection (`TriviaDisplayWindow.xaml`):

- High-fidelity full-screen monitor output featuring dynamic category headers, large question prompts, 4-color option cards with sequential opacity elimination fading, configurable 3-second warning timer badges, and live team leaderboard podiums.

// Edited on Aug 17, 2026 @ 12:31:30 -> Document single-song duet auto-clearing and session history persistence

Lyracist Pro & KSRotation Update - Duet / Backup Partner Support (August 2026)

Section: Singer Request Board & Rotation Management / Remote DJ & Mobile Portals

[Update Details]

1. Single-Song Duet Auto-Clearing & Session Reporting:

- When a performer finishes their song (`FinishSingerSong` or `AdvanceToNextSinger`), the duet partner is automatically archived into the session performance history database and then cleared from the active queue for their subsequent rounds.

- Preserves historical records for the entire night in `SongPerformance` (KSRotation), `PerformedSong` (Lyracist), Night Database snapshots, CSV exports (`Duet Partner` column), and PDF summary reports (`(with {PartnerName})`).

2. Mobile Patron Request Portal (`PatronPortal.html`):

- Added an optional "Duet / Backup Partner (Optional)" input field to the song submission form.

- Automatically displays for Karaoke requests and hides when Music (track playback) is selected.

- Forwards the partner's name as `duetPartner` with the request JSON payload.

- Rotation list displays `(with {PartnerName})` beside performer names.

3. Remote DJ Web Board (`dj.html`):

- Added "Duet / Backup Partner (Optional)" input field to the "Add Performer to Rotation" form.

- Added `(with {PartnerName})` badges to both the live queue cards and incoming request cards.

4. KSRotation Desktop (`MainWindow.xaml`):

- Added a dedicated "Duet Partner" editable column in the rotation queue grid.

- Incoming requests sidebar shows `(with {PartnerName})` for quick verification before accepting.

5. KSRotation.Maui (.NET MAUI):

- Added Duet Partner entry to the top Add Performer bar.

- Added Duet Partner field to the Edit Performer dialog.

- Displays `(with {PartnerName})` on rotation items and incoming requests.

6. Billboard Projection (`DisplayViewModel.cs`):

- Formats performer names as `{Singer} & {Partner}` on "Now Performing", "Up Next", and rotation crawls.

// Edited on Aug 17, 2026 @ 11:22:00 -> Document DJ QR Code enlarged popout modal and optical contrast enhancements

Lyracist Pro & KSRotation Update - DJ QR Code Enlarged Popout & Optical Contrast (August 2026)

Section: Singer Request Board / DJ Remote Control Portal

[Update Details]

1. Enlarged DJ Control QR Code Popout Modal:

- Clicking the DJ Control QR code on the main request panel opens a dedicated popout dialog (`DjQrCodePopoutWindow`).

- Features an extra-large (280x280) high-resolution QR code rendering with `NearestNeighbor` pixel scaling for easy optical scanning across the room.

- Prominently displays the active DJ Security PIN in large 24pt bold red font with a one-click "Copy PIN" button.

- Displays the full DJ web portal address with a one-click "Copy URL" button.

- Targeted Interaction: Popout activates only when the DJ Control Portal mode is active (`IsDjQrVisible`), leaving the singer song request QR code unaffected.

2. High-Contrast QR Code Optical Scanning:

- Standardized DJ and Wi-Fi QR codes to pure black on white (`[0, 0, 0]` on `[255, 255, 255]`) with a 21:1 contrast ratio, ensuring instant scanning on older tablet cameras (such as iPad Air).

- Added a red accent border (`#EF4444`, 2.5px) around the request panel container when DJ mode is active to clearly distinguish DJ mode.

// Edited on Aug 17, 2026 @ 09:55:00 -> Document Last Song priority banner display on all non-lyric screens

Lyracist Pro Update - Priority "Last Song" Banner on All Non-Lyric Screens (August 2026)

Section: Display & DJ Banners / Special Event Banners

[Update Details]

Added priority display handling for the "Last Song" event banner (`LastSong.png`):

- Highest Banner Precedence: When the "Last Song" event banner is selected or activated (via Settings, Karaoke Controls, or Remote DJ Board), it takes immediate priority over all other DJ and party banners.

- Multi-Screen Projection: The Last Song banner automatically projects on all non-lyric projection displays (the DJ Banner Screen and the Singer Rotation Billboard Screen), ensuring the entire venue sees the finale announcement.

- Uninterrupted Lyrics: The dedicated Lyric Projection screen continues rendering transparent CDG/scrolling lyrics without disruption so the final performer can complete their performance.

// Edited on Aug 15, 2026 @ 10:20:00 -> Document Session Performed Songs List & Textbox with 5-color rotation in Rotation Page

Lyracist Pro Update - Session Performed Songs History in Singer Rotation (August 2026)

Section: Singer Rotation / Session Performance History

[Update Details]

Added a dedicated Session Performed Songs column (Column 2) to the Singer Rotation page:

- Live Performance Tracking: Automatically records every completed song performance during the session, including performer name, song title, artist, key transposition, and timestamp.

- Dual Display Modes:

1. 5-Color Rotating List: Each performed song entry rotates sequentially through 5 distinct theme-tailored colors (Violet, Cyan, Emerald, Amber, Rose) in both Dark and Light color schemes for high visual clarity.

2. Session Textbox: Provides a clean multi-line read-only TextBox containing the complete text log of performed songs.

- Session Management: Includes 'Copy Session Songs to Clipboard' and 'Clear Session Performed Songs' quick-action buttons.

// Edited on Aug 15, 2026 @ 10:00:00 -> Document startup splash window bypass default

Lyracist Pro Update - Bypass Startup Splash Window (August 2026)

Section: Application Options / Startup Behavior

[Update Details]

The startup splash window pop-up has been disabled by default (`ShowSplashOnStartup = false`). Lyracist now launches directly into the main application window without displaying a splash screen or artificial loading delays.

Lyracist Pro Update - Title Bar Venue & DJ Info Block (August 2026)

Section: Main Application Window / Title Bar Integration

[Update Details]

Added dynamic Venue and DJ name indicators directly to the main window Title Bar:

- Placement: Positioned after the application title and before the pending request indicator bulbs.

- Stacked Format:

  - Top Line: `Venue: <name>` (bound to `VenueTitleText`, updating live when active venue changes in Settings).

  - Bottom Line: `DJ: <name>` (bound to `DjTitleText`, updating live when active DJ name changes in Settings).

Lyracist Pro Update - Settings Display Tab Controls & QR Code Overlay (August 2026)

Section: Settings Page / Display Tab / QR Code & Screen Controls

[Update Details]

Added explicit screen activation toggles and request QR Code overlay controls on the Display tab:

1. Screen Activation Checkboxes:

  - "Enable DJ Banner Screen": Toggles the DJ Banner projection window on/off (`IsDjBannerActive`).

  - "Enable Lyrics Projection Screen": Toggles the Lyrics projection window on/off (`IsLyricsActive`).

  - "Enable Singer Rotation Billboard Screen": Toggles the Rotation billboard projection window on/off (`IsRotationActive`).

2. Request QR Code Overlay Controls:

  - "Show request QR Code overlay on DJ Banner": Toggles rendering the patron request portal QR code overlay directly onto active DJ Banners (`IsDjBannerQrCodeEnabled`).

  - "Connect Instructions & QR Code Screen": Selects the target display screen/monitor for projected Wi-Fi and Song Request dual QR code banners (`ConnectInstructionsScreen`).

Lyracist Pro Update - Settings Display Tab (August 2026)

Section: Settings Page / Display Tab (4 Equal Columns)

[Update Details]

A dedicated "Display" tab has been added to the Settings screen in Lyracist. Displays & projection monitor assignments, DJ banner images, special event banner mappings, Star Wars crawl & spaceship overlay settings, and Connect & Request Instructions have been consolidated onto the Display screen.

1. 4 Equal Columns Grid Setup (referencing KsRotation layout):

- Column 1: Monitors & Screen Assignments (Lyrics Projection Screen, Singer Rotation Billboard Screen, DJ Banner Screen, Billboard View Mode, Mirror Lyrics).

- Column 2: DJ & Event Banners (DJ Banner Image upload/select/delete, Special Event Banners Mapping & Save button).

- Column 3: Star Wars Crawl & Spaceship Overlay (Crawl Intro Template & preview, Custom Crawl Text, Spaceship Overlay Font Size, Duration, Frequency, Snippets List & controls).

- Column 4: Connect & Request Instructions (Wi-Fi Password, Target Screen selector, Dynamic Banner Status info).

Lyracist Pro & KSRotation - Performer Portal Update (August 2026)

Section: Connect & Request Instructions Screen / Wi-Fi Password Manager

[Update Details]

A dedicated Connect & Request Instructions feature has been added to display scan-to-connect Wi-Fi and song request QR codes directly on projection monitors.

1. Scan-to-Connect Wi-Fi & Song Request Dual QR Code Banner:

- Automatically detects active Wi-Fi network SSID and song request portal URL.

- Generates a dynamic vector graphic (`ConnectInstructions.png`) featuring a WPA/WPA2/NoPass Wi-Fi scan-to-connect QR code (forest green `#064E3B`) and a song request portal QR code (pure black `#000000`).

- Clearly displays Wi-Fi network SSID, password status ("No Password Required" or explicit password), and portal URL.

2. Dynamic Target Screen Resolution Detection:

- Automatically detects the physical pixel dimensions (`width` x `height`) of the target display monitor (`1080p`, `1440p`, `4K 3840x2160`, etc.).

- Dynamically scales vector paths, fonts, card bounds, and QR code module density (`pixelsPerModule`) to guarantee razor-sharp 1:1 pixel rendering on any screen size.

3. Persistent Wi-Fi Password Store (`wifi_passwords.json`):

- Automatically remembers Wi-Fi passwords entered for specific SSIDs (venue Wi-Fi, travel router, mobile hotspot).

- Upon detecting a connected Wi-Fi network SSID, the application automatically recalls and populates the saved password into the Wi-Fi password input box.

- Entering or updating a password automatically updates `wifi_passwords.json` for that SSID.

4. Dedicated Green GroupBox Layout & Screen Selector:

- Located in the Display settings tab inside a dedicated Green GroupBox styled under Column 1 (directly under Display & Projection).

- Features a dedicated **Target Screen / Monitor** selection dropdown with options: "None", "All Screens / Monitors", or individual connected display monitors.

5. Integrated Help System Topic Expansion:

- Added dedicated 'Connect & Wi-Fi Instructions' topic in KSRotation and '15. Connect & Request Instructions Screen' topic in Lyracist (`HelpViewModel.cs`), providing step-by-step guidance for dual QR code setup, password management, resolution scaling, and target display selection.

Section: Singer Request Board / Patron Request Portal

[Update Details]

A new online search capability has been added directly to the patron request portal.

1. Online Song & Artist Lookup:

- When users access the Patron Request Portal via QR Code, they will see a new field labeled "Search Songs & Artists (Online)".

- Type query keywords (such as the song name or artist) to query the iTunes search index in real-time.

- Tap any suggestion from the search results list to automatically populate the "Song Title" and "Artist Name" fields in the form.

2. Connection Dependency:

- This feature runs on the patron's mobile browser, using the patron's own internet connection (via mobile data or internet-enabled Wi-Fi).

- The host system (main KJ laptop/tablet) does NOT require internet or external drives connected for this search to function.

3. Graceful Offline Fallback:

- If the patron's phone has no internet connection (e.g. no cellular signal or local-only hotspot), the search bar will gracefully show a "Search unavailable (offline)" message.

- The manual entry form remains fully functional under all network conditions.

Section: Display Projection Window / Display Monitor Selection

[Update Details]

Operators can now select a specific monitor for displaying the rotation projection window (Singer Display Window) from a dropdown list of available system screens.

1. Target Monitor Dropdown:

- Located in the "Settings" tab under a new group titled "Display Monitor".

- Displays all connected monitors with their indices, bounds, and whether they are primary.

- When selected, the display window is repositioned on that specific screen.

- If the selected monitor is unplugged or unavailable, the application falls back automatically to the secondary screen if available, or the primary screen.

2. Dynamic Window Relocation:

- Selecting a different monitor dynamically moves the active, maximized display window to the target screen immediately without needing to close and re-enable it.

3. Refresh Monitors List:

- If a new monitor is connected while the application is running, operators can click the "Refresh" button next to the dropdown to scan and reload the system's screen configuration.

Section: Rotation Management / Paused and Inactive Singer Controls

[Update Details]

Updates have been made to both the main console and the DJ web portal to manage paused and deleted (inactive) singers more effectively.

1. Paused Singers:

- Operators can pause a singer using the Pause button in the console or the remote DJ portal.

- Paused singers retain their index/position in the rotation list, but their row is grayed out.

- Paused singers cannot be marked as the current singer or next singer until they are explicitly unpaused (resumed). They are automatically skipped in the rotation.

2. Deleted/Inactive Singers:

- Clicking the Delete (trash can) button no longer deletes a singer from the database. Instead, it marks them as inactive (deleted) and moves them to the end of the rotation list.

- Inactive singers' rows are grayed out.

- When a singer is inactive, their delete button icon changes to a "↺" (Restore) button in the console and "Restore" button in the DJ portal.

- Clicking Restore reactivates the singer and moves them back to the end of the active section in the rotation.

Section: Patron Portal / Background Music & Song Requests

[Update Details]

Patrons can now request background music/song tracks to be played in between karaoke performances directly from the web portal.

1. Karaoke vs. Music Selection:

- On the Patron Request Portal, users will see a new selector under their name: "Karaoke (Sing)" and "Music (Play)".

- Selecting "Music" limits the submission to one song, hides the "Add another song" option, and marks the request category as Music.

2. DJ Incoming Requests Queue:

- On the DJ screen and the remote DJ portal web interface, incoming music requests are labeled with a green [MUSIC] badge, distinguishing them from standard purple [KARAOKE] requests.

- Standard duplicate logic is updated so a singer can have one active karaoke slot and one background music slot simultaneously.

3. Rotation List Layout:

- Accepting a music request places it in the active rotation queue.

- For music entries in the queue, the 10 round checkboxes are automatically hidden and replaced by a "♪ Background Music Track" label.

- On the DJ screen, the name, song, and artist fields of music tracks are highlighted in bold green for easier identification.

4. Projection Display Support:

- Music requests show up with a "[MUSIC]" prefix in the active list and flip tile board.

- When a music request is active/next:

  - The projection window's Vegas Marquee style shows "★ BACKGROUND MUSIC ★".

  - The Vinyl Turntable style displays "♪ NOW PLAYING".

- The Star Wars Crawl renders the entry with a "NOW PLAYING" header, displaying the song title as the primary text, followed by the artist and requester details below.

Section: Rotation Management / FlipTile Board View Removal

[Update Details]

The obsolete "Flip Tile Board" view option has been removed from all applications (including KSRotation and Lyracist).

1. Interface Clean-up:

- The "Flip Tile Board" selection option has been removed from the Projection View dropdown menus.

- All underlying split-flap character structures and visual board panels have been completely removed from the singer display screen to simplify the interface and optimize performance.

Section: Main Console / Active Venue & DJ Title Bar Integration

[Update Details]

To resolve screen space and layout wrapping issues on 1920x1080 laptop screens, the active Venue and DJ name displays have been integrated into the standard window title bar.

1. Dynamic Title Bar Display:

- The active Venue and DJ name are now visible in the title bar of the main console window, e.g. "Karaoke Singer Rotation - Venue: [VenueName] | DJ: [DjName]".

- The title updates dynamically in real-time when selecting or updating the active Venue or DJ under Settings.

2. Unwrapped Control Bar:

- The status badge has been removed from the main rotation tab toolbar, ensuring that other controls (such as checkboxes and view selectors) have maximum screen width and do not wrap layout rows.

Section: Mobile & Desktop App / Standardized Splash Screen

[Update Details]

The splash screen for the mobile and desktop versions of Karaoke Singer Rotation (KSRotation.Maui) has been redesigned and standardized using cross-platform vector assets.

1. Vector Centerpiece Logo:

- Features a clean, high-contrast microphone centerpiece surrounded by vibrant purple and orange rotating orbital paths.

- Simplified to ensure 100% compatibility with standard .NET MAUI Resizetizer compilation tools on both Android and Windows.

2. Obsidian Dark Theme Background:

- The launch background has been updated to a deep premium obsidian color (#12091A) on both platforms, providing a seamless visual transition during startup.

Section: Display Projection Window / DJ Banner and Rotation Screen Collision Priority

[Update Details]

To ensure proper projection setup, priority logic has been added when both the DJ Banner and the Rotation Display target the same screen/monitor.

1. Same-Screen Deconfliction:

- The DJ Banner will usually be displayed on the screen while everything is getting setup and the list is started.

- Once the Rotation Display is enabled (or moved) to the same screen, the DJ Banner is automatically disabled and hidden so the rotation list can display.

2. Multi-Screen Independent Operation:

- If the DJ Banner and the Rotation Display target different screens, they will both be displayed and operate simultaneously.

Section: Main Console / Front Page Music Search & Rotation Insertion

[Update Details]

Operators can now search the background music library directly from the main Karaoke dashboard, add tracks to the rotation queue as music, and configure automated playback.

1. Music Library Search Tab:

- A new "Music Library" tab has been added to the search panel on the front page of Lyracist, positioned next to the "Local Library" tab.

- Searching via the main search bar queries both the karaoke library and the background music library (tracks where IsKaraoke is false) simultaneously.

2. Insertion into Performer Rotation:

- Displays search results with Song and Artist columns.

- Click the "Add to Rotation" (+) button or double-click a song in the list to immediately insert the track into the active rotation queue as a music track (highlighted in green and labeled "♪ Background Music Track").

- Music tracks are added under the generic name "Music Request", bypassing the duplicate check and database persistence, allowing you to insert multiple independent music tracks into different positions in the queue.

- This provides a convenient mechanism for hosts to interject background music tracks into the main rotation occasionally without requiring a patron request.

3. Auto-Play & Configurable Delay:

- Added a new control row right below the queue filters on the front page containing an "Auto-Play Rotation Music" checkbox and a "Delay" selector.

- When checked, background music tracks in the rotation queue will automatically start playing after the previous singer finishes.

- The delay selector allows customizing how many seconds (0s to 30s) the system waits before starting the music track.

4. Automatic Queue Cleanup:

- Once a background music track finishes playing and is marked Done, it is automatically removed from the rotation queue rather than remaining for subsequent rounds.

Section: Display Settings & Remote DJ Board / Special Event Banners

[Update Details]

Special Event Banners have been introduced to Lyracist, KSRotation, and the remote DJ Board app. These banners share the same screen space as the standard DJ banners, allowing operators and DJs to toggle custom banners for special occasions.

1. Pre-Configured Events:

- Includes default, high-quality event graphics for "Birthday", "Wedding", "Engagement", and "Anniversary".

2. Configurable Mapping:

- In both Lyracist and KSRotation Settings panels, operators can define custom Event Names and map them to image/video files residing in the "EventBanners" directory.

3. Host Control Toggles:

- A row of radio buttons has been added directly to Lyracist's "Karaoke Page" (under DJ Banner Selection) and KSRotation's "Display" tab (under DJ & Special Event Banners).

- Selecting an event name immediately projects the mapped special event banner on the projection screen. Selecting "None" reverts the screen to the standard DJ banner.

4. Remote DJ App Integration:

- The web-based remote DJ dashboard (`dj.html`) has a new "Special Event Banners" panel.

- Polling queries the active event state from the host. Tapping any event button (or "None") updates the host display in real-time, allowing DJs to easily manage special event banners from their phone/tablet.

Section: Display Settings & Remote DJ Board / Special Event Banners Management & Styling

[Update Details]

Updates have been made to Special Event Banner management in KSRotation and the remote DJ Board app, along with high-contrast styling and optical scanning enhancements.

1. Upload, Add & Delete Special Event Controls:

- Added "Upload Banner", "Add Event", and "Delete" buttons to the Special Event Banners Mapping panel under KSRotation's Display tab.

- Uploading copies banner graphics directly to the EventBanners directory.

2. Standard Pre-Saved Event Protection & Display:

- The 4 standard pre-saved events ("Birthday", "Wedding", "Engagement", "Anniversary") are guaranteed to exist with auto-generated graphics if missing.

- Standard events are protected from accidental deletion.

- The mapping grid displays standard events as clean fixed labels, reserving drop-down list boxes and edit inputs exclusively for custom events.

3. Dual-Theme Dynamic Purple & Light Lavender UI Styling & Dark Maroon QR Code:

- Button text, borders, and RadioButton labels dynamically adapt based on theme mode: bright light lavender (#E9D5FF text, #C4B5FD border) for dark backgrounds, and rich deep purple (#6D28D9 text, #7C3AED border) for light backgrounds to ensure high legibility across all theme modes.

- The DJ QR Code foreground color has been updated from bright crimson to dark maroon/brick-red (RGB 128, 0, 32 / #800020), ensuring high optical contrast against white tiles for scanning by all mobile cameras.

4. 11+ Performance Rotation Advance:

- Clicking Finished Song in KSRotation for a performer who has completed all 10 round checkboxes now seamlessly advances the rotation, promotes queued songs, and logs performance details to the database history.

Section: Rotation Advancement & Default Event Banners Update

[Update Details]

Updates have been made to rotation sequence advancement and standard default event banners in KSRotation and Lyracist.

1. Robust 1-Click Rotation Advancement & Rollover:

- When marking a singer finished (via "Done", "✓", or remote DJ portal), rotation sequence advancement calculates the next active singer sequentially starting after the finished singer's index.

- When the last singer in the rotation finishes, rotation automatically rolls over to the top performer in 1 click.

- Rotation advances reliably even if the finishing singer was not marked as current prior to clicking "Done".

2. "Last Song" Default Event Banner:

- Added "Last Song" (`LastSong.png`) to the standard pre-saved event banners list alongside Birthday, Wedding, Engagement, and Anniversary.

- Standard pre-saved events are protected from deletion and automatically mapped to their respective graphics.

Section: Lyracist Live Trivia / In-App Help, About Tabs & Authoritative Timer Controls

[Update Details]

Dedicated Help and About systems have been added to the LyracistTrivia host application, and help topics across Lyracist and KSRotation have been updated with full trivia operations, pre-game lobby features, and timer rules.

1. LyracistTrivia "Help" and "About" Tabs:

- **Help Tab (` ❓ Help `)**: Features a split-pane interactive index menu with 7 comprehensive documentation topics:

  * ` 🎯 1. Game Master Command Deck `: Pack loading, 1-click pre-game lobby launch, skip countdown, auto-advance loop, manual reveals, and live response visualizer.

  * ` 📺 2. 70:30 Pre-Game Lobby & 16:9 Banners `: Widescreen 70:30 projection layout, 16:9 category banners, countdown clock, stacked Wi-Fi & Mobile Join QR codes, bottom connection bar, and scrolling ticker.

  * ` ⚙️ 3. Settings & Timer Override Rules `: Game Master Question Answer Window timer override, 5s wrong answer elimination, 5s reveal buffer, 1,000 base points, +500 speed bonus, +10% streak multiplier, and wrong answer deductions.

  * ` 📱 4. Mobile Buzzer Web App `: Zero-install player buzzer portal (` http://<LAN-IP>:8085/trivia `), player/team registration, 4-button color buzzer, haptic feedback, real-time elimination fading, and personal ranking.

  * ` 👥 5. Players & Teams Management `: Real-time roster table, score tracking, answer streaks, accuracy stats, and instant player kick moderation.

  * ` 🗄️ 6. 14 Category Packs (2,100 Questions) `: 14 curated packs (150 questions each), SQLite database seeding, and custom JSON packs.

  * ` 🖥️ 7. Multi-Monitor & Shortcut Keys `: Multi-monitor selection, Esc / floating close button, F11 fullscreen, and DJ banner auto-pause deconfliction.

- **About Tab (` ℹ️ About `)**: Displays application logo, version info (` v26.8.19.20 `), company (` PAROLE Software `), author (` Dennis N. Maidon `), copyright, and system specifications summary.

2. Authoritative Question Answer Window Timer Rule:

- The Game Master's configured Question Answer Window in Settings (`DefaultQuestionSeconds`, default: 15s) is authoritative for all live rounds.

- If the Game Master changes the timer to 10s, 20s, 25s, 30s, etc., the game engine executes every question using the Game Master's custom duration, overriding the 15s default written in the JSON files.

3. Cross-Application Help System Updates:

- **Lyracist Help Screen (`HelpViewModel.cs`)**: Added Topics 16 ("Lyracist Live Trivia"), 17 ("Trivia Settings, Timers & Scoring Rules"), and 18 ("Trivia JSON Database Schema & Custom Pack Guide").

- **KSRotation Help Screen (`MainWindow.xaml`)**: Updated "🎯 Trivia Night Pro" topic with 70:30 pre-game lobby, 16:9 category banners, 14 categories (2,100 questions), timer override rules, and custom JSON packs.

Section: Trivia Database Creator & Pack Studio (`TriviaDbCreator.exe`)

[Update Details]

A dedicated standalone question authoring and category management studio (`TriviaDbCreator.exe`) has been introduced to the Lyracist suite.

1. Fluent UI / MVVM Architecture:

- Built with WPF .NET 10, CommunityToolkit.Mvvm, and WPF-UI (Fluent Dark Theme / Mica styling).

- Real-time dirty state tracking, auto-save validation, and clean navigation.

2. Pack & Question Management Features:

- **Left Sidebar Pack Explorer**: Auto-discovers and navigates all JSON question packs in `TriviaData/packs/` with search filtering and question count indicators.

- **Pack Metadata Editor**: Configure Pack Title, ID slug, Category, and Theme Description.

- **Center Panel Questions Table**: Filterable table listing Question #, ID, Prompt, Difficulty badge (`Easy`/`Medium`/`Hard`), and Correct Option indicator (`▲ A`, `◆ B`, `● C`, `■ D`).

- **Right Panel Question Editor**: Edit question prompt, 4 distinct multiple-choice options (A, B, C, D), radio selector for correct answer, difficulty rating, and reveal explanation note.

- **1-Click Option Balancing**: Automatically shuffles option positions across all questions in a pack to guarantee an even ~25% distribution across choices A, B, C, and D without altering question accuracy.

- **16:9 Banner Studio**: Generates matching high-resolution 16:9 Category Announcement Banners (`TriviaData/Banners/{pack}.png`) with one click.

- **Direct SQLite Seeding**: 1-click database synchronization updating `TriviaData/trivia.db`.

3. In-App Help & Documentation System (`❓ Help & Instructions`):

- Built-in split-pane documentation browser covering 8 in-depth topics:

  * **1. Quick Start & Overview**: General workflow and 3-column workspace orientation.

  * **2. Creating & Managing Databases**: How to add, configure, rename, and delete category packs.

  * **3. Adding, Editing & Duplicating Questions**: Multi-choice option inputs, radio toggles for correct answer, difficulty ratings, and reveal explanations.

  * **4. Balancing Answer Distributions (25% Rule)**: Real-time distribution analysis and 1-click option shuffling.

  * **5. 16:9 Category Announcement Banners**: Widescreen banner rendering into `TriviaData/Banners/`.

  * **6. SQLite Database Sync & Live Hosting**: Cross-application sharing with `LyracistTrivia`, `Lyracist`, and `KSRotation`.

  * **7. Importing & Exporting Databases**: Backup, restore, and pack exchange.

* **8. JSON Database Format & Schema Reference**: Syntax rules, sample JSON code, and field specifications.

- ** ℹ️ About Tab**: Displays official brand logo, version, company, author, copyright, and technology stack.

Section: Full Trivia System Alignment & Screen Parity Across Solution Applications (Lyracist, KSRotation, Lyracist.Trivia)

[Update Details]

Complete visual, architectural, and operational feature parity has been implemented across the embedded Trivia engines in Lyracist and KSRotation to match the standalone Lyracist.Trivia Game Master application. Furthermore, strict screen display prioritization and automatic pause deconfliction ensure that live Trivia seamlessly integrates around primary karaoke singer performance and rotation workflows.

1. 16:9 Big Screen Projection Window (`TriviaDisplayWindow`) in Lyracist & KSRotation:

- **Unified 70:30 Pre-Game Lobby Layout**:

* **70% Left Hero Column**: Full widescreen showcase displaying high-resolution 16:9 Category Announcement Banners (`TriviaData/Banners/{pack}.png`) or real-time generated collage banners for multi-pack mixed rounds (`TriviaBannerGenerator`). Includes category theme title, subtitle, and ambient illumination.

* **30% Right Onboarding Stack**:

1. ** ⏱️ Pre-Game Countdown Clock**: Digital countdown timer displaying time remaining until round start.

2. ** 📶 1. Connect to Wi-Fi QR Card**: Scan-to-connect Wi-Fi QR code with venue SSID and WPA password for instant patron phone connection.

3. ** 📱 2. Join Trivia Game QR Card**: Scan-to-join mobile buzzer QR code pointing to `http://<LAN-IP>:8085/trivia`.

* **Bottom Connection & Copyright Bar**: Displays mobile buzzer URL, company copyright (`© 2026 PAROLE Software`), and app branding.

* **Top Scrolling Marquee Bar**: Continuous horizontal ticker cycling venue announcements, host branding, game rules, and top scorer leaderboards with crown indicators.

- **Active Gameplay & Question Presentation**:

* Scaled question prompt cards with automatic Viewbox text fitting.

* 2x2 stylized answer option cards (▲ Red/Purple, ◆ Cyan/Blue, ● Amber, ■ Rose/Green) with real-time wrong option elimination fading and correct answer highlights upon reveal.

- **Post-Game Celebration & Intermission**:

* Winner trophy fanfare with individual champion score or winning team roster breakdown.

* Live intermission timer countdown banner indicating when the next round will auto-start.

2. 3-Column Game Master Console Deck (Lyracist `TriviaPage.xaml` & KSRotation `MainWindow.xaml`):

- **Header Bar**: Active venue indicator, live connected player count badge, target monitor projection dropdown, and 1-click "📺 Open TV Display" / "📡 Open Big Screen" button.

- **Left Column (Game Master Controls)**:

* Category Pack Checklist (check 2+ to mix categories).

* Questions Per Game preset selector (5, 10, 15, 20, 25, 50, 100 questions).

* Auto-Run Game switch (15s answer, 5s fade, 5s reveal).

* 1-Click `"🎯 Launch Pre-Game Countdown"` and `"▶ Start Game Now (Skip Countdown)"`.

* Live Question Controls (Start Question, Pause/Resume, Lock & Reveal Answer, Next Question, Reset Session).

* Mobile Web Server status and direct join link.

- **Center Column (Live Round & Answer Distribution Visualizer)**:

  * Active Round header and ticking countdown timer.

  * Current Question prompt card.

  * Real-time **Patron Answer Distribution Visualizer** displaying live vote counts, percentages, and progress bars for options A, B, C, and D.

  * 2x2 Answer Option previews.

- **Right Column (Live Player Leaderboard & Moderation)**:

  * Live player roster sorted by total score.

  * Player name, registered team name, and current score.

  * Individual player kick action (`✕`) for removing disruptive participants.

3. Dedicated Separate Trivia Settings (Lyracist `TriviaSettingsPage.xaml` & KSRotation `MainWindow.xaml`):

- **Question Timers & Flow**: Time limit (s), Warning countdown (s), Wrong option elimination fade (s), Post-reveal delay (s), and Auto-advance toggle.

- **Scoring & Multipliers**: Base points per question (1,000 pts), Wrong answer deduction (0 pts default), Speed bonus toggle & maximum (+500 pts), Streak bonus toggle & multiplier (+10%/streak).

- **Venue & Host Customization**: Active venue name, Host/DJ name, and customizable Pre-Game Welcome Banner template supporting `{venue}` and `{dj}` replacement tokens.

- **Intermission & Pre-Game Controls**: Auto-start next game toggle & delay in minutes, Pre-game countdown delay in minutes & auto-start toggle upon timer expiry.

- **Local Network & Wi-Fi Access**: Server port (8085 default), Wi-Fi SSID, Wi-Fi password, and 1-click `"Auto-Detect Wi-Fi"` tool.

- **Target Projection Monitor**: Selectable display monitor with true physical DPI awareness.

- **Save & Reset**: 1-click Save Settings and Reset to Defaults with animated status toasts.

4. Karaoke & Rotation Screen Priority / Automatic Pause Deconfliction:

- In both Lyracist and KSRotation, Trivia operates as a secondary entertainment module that automatically respects and yields to primary karaoke and singer rotation functions:

* **Priority 1**: Karaoke Lyrics Projection Window during song playback.

* **Priority 2**: Singer Rotation Queue Display between songs.

* **Priority 3**: DJ / Venue Announcement Banners.

* **Priority 4**: Lyracist Live Trivia Projection.

- When a singer performs or the rotation screen is cast, the Trivia engine automatically pauses with a clear pause status (`"Karaoke Performance"` or `"Rotation Screen Active"`) and hides or yields the display.

- Once song playback or rotation display concludes, Trivia automatically resumes seamlessly.

Section: Live Trivia Scoring / Tiered Option Value Multiplier (100% / 70% / 40%)

[Update Details]

A dynamic tiered scoring system has been integrated into Lyracist, KSRotation, and Lyracist.Trivia, tying point awards directly to the visual option elimination phases during live question play.

1. Progressive Option Elimination & Value Tiers:

- **4 Options Visible (0 Eliminated)**: 100% of Base Points (1,000 pts default). Rewarding early, confident buzz-ins before any wrong answers fade.

- **3 Options Visible (1 Eliminated)**: 70% of Base Points (700 pts default). Occurs when the first wrong answer fades at the 2/3 countdown interval.

- **2 Options Visible (2 Eliminated / 50-50)**: 40% of Base Points (400 pts default). Occurs when the second wrong answer fades at the 1/3 countdown interval.

- **Expected Value Balance**: Discourages patrons from blindly waiting for the 50/50, ensuring early knowledge is awarded a decisive 2.5× advantage (1,000 vs 400 pts) while casual patrons can still score points.

2. Mobile Buzzer Web App (`trivia.html`):

- Real-time Point Multiplier Pill (`#value-multiplier-pill`) prominently displayed above the buzzer grid.

- Dynamically updates as options fade: `⚡ 100% VALUE (1,000 pts)`, `⚡ 70% VALUE (700 pts)`, `⚡ 40% VALUE (400 pts)`.

- Wrong options fade and disable on the phone in real time as they fade on the big screen.

3. Trivia Settings & Configuration:

- Toggle checkbox: "Tiered Option Value (4=100%, 3=70%, 2=40%)"

- Customizable percentage fields for 4 Opt (%), 3 Opt (%), and 2 Opt (%) across Lyracist (`TriviaSettingsPage.xaml`), KSRotation (`MainWindow.xaml`), and Lyracist.Trivia (`MainWindow.xaml`).

Section: Lyracist / Per-Singer Key, Tempo, Mic Level & Vocal EQ Recall

[Per-Singer Audio & Performance Memory]

1. Automatic Key & Tempo Memory per Performer per Song:

- When a singer is added to the rotation (via local library search, patron kiosk request, or manual queue entry), Lyracist automatically checks the singer's performance history (SingerHistory).

- If the singer has previously performed that specific song, their saved Key transposition (semitone offset) and playback Tempo (speed multiplier, e.g., 1.1x) are automatically recalled and applied to the rotation entry.

- If the song has not been sung by that performer before, Lyracist falls back to the performer's general profile preferences (SingerAudioSettings), ensuring personalized pitch and speed defaults are maintained without manual host intervention.

- When the performer takes the stage and their track begins playback, the live Pitch Transpose and Speed sliders in the Live Deck reflect these recalled settings immediately.

2. Per-Singer Mic Level & Vocal EQ Recall:

- When a performer's turn begins and their track starts, Lyracist automatically loads and applies the performer's saved vocal gain (Gain), 3-band parametric EQ (Treble, Mid, Bass), dynamic compressor threshold, and peak limiter from SingerAudioSettings.

- The on-screen Audio Controls sliders in the Karaoke dashboard update in real time to reflect the active performer's profile, delivering consistent microphone leveling and tonal balance throughout the show.

3. Quick Action Buttons in Audio Controls ("Save to Singer" & "Recall Singer"):

- Located on the Audio Controls card header alongside the "Reset Audio" button:

* "Save to Singer": Instantly saves the current Audio Controls slider values (Volume as Vocal Gain, Treble, Mid, Bass, Compressor, Limiter) as the active singer's saved defaults in SingerAudioSettings, and updates their performance history with the current Pitch and Speed settings.

* "Recall Singer": Re-reads and reapplies the active singer's saved audio defaults from the database on demand, resetting the DSP chain and slider controls to their stored preferences if adjustments were made during the song.

4. Singer History Key & Speed Tracking:

- The Singer History table in the Karaoke tab now includes dedicated "Key" and "Speed" columns displaying past transposition and tempo settings for every completed performance.

- Re-adding a song from Singer History back into the rotation queue preserves and applies the historical Key and Tempo settings seamlessly.

Section: Lyracist & KSRotation / Session Song Tracking, Duplicate Blocking, Users Management & Performer Selfies

[Session Song Tracking & Duplicate Song Blocking]

1. Session-Wide Song History Tracking:

- Tracks all songs performed and queued during the active show session in both Lyracist and KSRotation.

- Dedicated "Session Performed Songs" list on the Rotation tab displays order of performance, singer, duet partner, song title, artist, key transposition, and speed multiplier.

- Actions to copy the session song list to clipboard or clear the session history when beginning a new night.

2. DJ Setting - Block Duplicate Songs in Session:

- Configurable checkbox in Settings > Rotation Settings in both Lyracist and KSRotation (BlockDuplicateSongsInSession).

- When enabled, patrons submitting song requests via the web patron portal cannot request a song that has already been sung or is currently queued in rotation during that session.

- Patrons receive an immediate, polite notice: "[Song Title] has already been performed or queued in this session. Duplicate songs are blocked by the DJ."

[Dedicated "Users" Navigation Tab & Singer Management]

1. Full Performer Directory & Editor ("Users" Tab):

- New primary navigation tab in Lyracist sidebar with Person icon.

- 2-pane master-detail interface: searchable list of all registered performers with instant filter by name or email, level badges, XP scores, and song counts.

- Create new singer accounts or delete obsolete accounts.

2. Performer Profile & Credentials Editor:

- Edit Singer Name, 4-digit Patron Portal PIN code, Email, Vocal Range (Soprano, Mezzo-Soprano, Contralto, Countertenor, Tenor, Baritone, Bass), Custom Title / Stage Nickname, Experience Score (XP), and DJ Notes.

3. Per-Singer Vocal & Audio Defaults:

- Dedicated tab for audio preferences: Microphone Gain (Volume), Key Transposition (-12 to +12 semitones), Playback Speed (0.8x to 1.2x), 3-Band Parametric EQ (Treble, Mid, Bass), Dynamics (Compressor, Limiter), and Sound Check Remarks.

- Settings automatically preload whenever the performer is queued or takes the stage.

4. Comprehensive Performance History Record:

- "Performance History" tab displaying a complete historical record of every song sung by the performer.

- Columns for Song Title, Artist, Key Transposition, Playback Speed (Tempo), Source (Local, YouTube, etc.), and Date/Time stamp.

- Direct Actions: "Queue" button instantly re-queues the past performance back into the live rotation preserving the performer's exact key and tempo settings; "Remove" button purges individual entries.

5. Merge Duplicate Singer Accounts:

- "Merge Duplicate..." button opens a dedicated consolidation modal.

- Selects another singer profile from a dropdown list to consolidate into the primary account.

- Transfers all historical song performances, past requests, active rotation entries, audio profile defaults, and XP points to the target account, then permanently removes the duplicate account.

[Performer Selfie / Photo Uploads & Billboard/Turntable Display]

1. Patron Portal Selfie & Photo Uploads:

- Patrons can upload a selfie or profile image directly through the smartphone patron portal (/api/singer/avatar/upload).

- Safe validation and storage in Data/Avatars/ with database record linking.

- DJs can view, upload, or clear photos in the desktop "Users" tab.

2. Vegas Billboard ("Vegas Marquee") Display:

- On the Broadway/Vegas marquee rotation screen, a high-resolution circular framed photo with gold neon glow effect is prominently rendered right above the singer's name on the big screen.

3. Vinyl Record ("Now Spinning") Turntable Display:

- On the turntable rotation screen, the circular framed photo is showcased beside the "â™ª NOW SPINNING" banner and singer title while the vinyl record spins.

Section: Lyracist & KSRotation / Session Start/Stop Scheduling & Last Request Cutoff

[Session Start/Stop Scheduling & Request Cutoff]

1. Session Schedule Management:

- DJs can configure exact start and stop times for the active karaoke session via a dedicated "Session Schedule & Request Cutoff" settings groupbox in both Lyracist and KSRotation.

- Supports 30-minute dropdown interval selection or custom freeform time entry with full 12-hour AM/PM and 24-hour military format compatibility.

- Full overnight support crossing midnight (e.g., 8:00 PM to 2:00 AM, 9:00 PM to 1:30 AM).

- "Enable Session Schedule" toggle enforces session boundaries: patron portal and kiosk submissions attempted before the session starts or after it concludes are blocked with an informative message informing patrons of the scheduled hours.

2. Last Request Cutoff Time:

- Configurable "Enable Last Request Cutoff" toggle and "Last Request Time" picker (default 1:30 AM).

- Once the cutoff time is reached, patrons and kiosk users can no longer submit new song requests for the remainder of the night.

- Patrons receive an immediate, polite notice: "Song requests are now closed for tonight. The cutoff time for requests was [Time]."

- Works seamlessly alongside session start/stop scheduling or independently for open-ended shows.

Section: KSRotation / Dedicated Users & Singer Account Management Tab

[KSRotation Users Tab & Singer Account Management]

1. Dedicated "Users" Navigation Tab in KSRotation:

- KSRotation now features a dedicated top-level "Users" tab (located directly between Settings and Trivia) providing full feature parity with Lyracist for performer account administration.

- Master-detail interface: Left pane features real-time search filtering across singer names, emails, and custom titles, complete with circular avatar thumbnails, singer song counts, vocal ranges, and level badges.

- Performer lifecycle: Instant "New Singer" creation and "Delete Singer" options directly from the master list.

2. Performer Profile & PIN Credentials Editor:

- Full editing of singer name, 4-digit patron portal PIN code, email address, vocal range dropdown selection, custom stage titles / nicknames, experience score (XP), and DJ performer notes.

- Automatic synchronization with the shared SQLite database (Data/lyracist.db), ensuring logins, kiosk access, and performer credentials remain identical across both applications.

3. Vocal & Audio Profile Defaults:

- Dedicated "Audio Defaults" sub-tab to configure preloaded settings: Microphone Gain, Key Transposition, Playback Speed (tempo), 3-Band Parametric EQ (Treble, Mid, Bass), Compressor Amount, Limiter Threshold, and Sound Check Remarks.

- Settings persist per performer and are automatically recalled whenever the performer sings.

4. Performance History & One-Click Rotation Queuing:

- Dedicated "Performance History" sub-tab logs all historical songs performed by the singer.

- Displays Song Title, Artist, Key Transposition, Playback Speed, Source, and Timestamp.

- Includes one-click "Queue" action to immediately insert a past favorite song into the active rotation round, and a "Remove" button to delete outdated records.

5. Duplicate Performer Account Consolidation:

- "Merge Duplicate..." button opens a dedicated modal overlay.

- Selects any duplicate performer profile from a dropdown list to consolidate into the currently selected primary account.

- Automatically transfers all song performance history, active queue items, and accumulated XP points to the primary account, then permanently removes the duplicate profile.

6. Performer Photo & Avatar Management:

- Dedicated avatar controls with "Upload Photo" and "Clear Photo" actions.

- Uploaded photos are stored in Data/Avatars/ and immediately reflected in rotation billboards, marquee screens, and the patron portal.

Section: Users & Singer Management / Webcam Performer Photo Capture (Lyracist & KSRotation)

[Webcam Performer Photo Capture & Avatar Attachment]

1. DJ Live Webcam Photo Capture ("Take Photo"):

- A prominent "Take Photo" button is now available in the Performer Profile Header card on the Users tab across both Lyracist and KSRotation.

- Allows the host/DJ to capture a live photo of the performer using any connected USB webcam, integrated laptop camera, or external video capture device.

2. Real-Time Camera Preview & Headshot Framing:

- Clicking "Take Photo" launches a modern, dimmed modal dialog ("Webcam Performer Snapshot").

- Camera Selector: Dropdown automatically discovers and lists all connected video capture devices, allowing instant switching between multiple cameras (e.g., laptop internal webcam vs. DJ desk USB camera).

- Viewport: Smooth live video feed with an intuitive dashed circular headshot framing guideline to ensure performer portraits are centered and flattering.

3. Snapshot Capture, Retake, and Square Aspect Ratio Cropping:

- "Capture Snapshot": Freezes the live frame instantly for DJ and performer review.

- "Retake": Discards the preview and returns immediately to the live stream if the performer blinked or wishes to pose again.

- "Save & Apply Photo": Automatically centers, square-crops, scales, and encodes the image as a high-quality JPEG into the Data/Avatars/ folder.

- The captured photo is immediately linked to the performer's profile and displayed across rotation marquee screens, the "Now Spinning" vinyl turntable banner, and the patron mobile web portal.

4. Comprehensive Image Attachment Options:

- "Take Photo": Live webcam snapshot directly from the DJ console.

- "Upload Photo": File picker to attach existing image files (JPEG, PNG, WEBP) from disk.

- "Clear Photo": Resets the performer back to the default avatar icon.

Section: Karaoke Store & Licensed Track Import (Lyracist ONLY)

[Overview & Purpose]

The "Store" tab is a dedicated, top-level navigation view exclusive to the main Lyracist application. It provides a seamless search, purchase, and automatic import workflow for

licensed karaoke tracks from commercial providers (such as Karaoke Version and Party Tyme) without web scraping or unauthorized audio proxying.

[Deep-Link Store Search]

1. Search Bar & Provider Querying:

- Enter any song title, artist, or keywords in the search box.

- "Search Karaoke Version": Opens your default web browser directly to Karaoke-Version.com search results for your query.

- "Search Party Tyme": Opens your default web browser directly to PartyTyme.net search results for your query.

- Pressing Enter inside the search box automatically launches Karaoke Version search.

2. Legitimate Purchasing:

- Hosts purchase licensed tracks directly on the manufacturer's secure website and download their files (MP3+G, CDG, MP3, ZIP, or MP4) locally.

[Purchased Tracks Folder Watcher & Auto-Import]

1. Purchased Tracks Folder:

- Specifies the directory monitored for new track downloads (defaults to the user's Downloads folder).

- "Browse..." and "Open in Explorer" buttons provide quick path selection and direct file access.

2. Automatic Background Watcher:

- "Automatically import new files from Purchased Tracks folder" toggle switch enables an asynchronous FileSystemWatcher.

- Automatically detects incoming .mp3, .cdg, .zip, and .mp4 files.

- Includes intelligent write debouncing and file-lock verification to ensure browser downloads are 100% complete before import begins.

- Coalesces matching .mp3 and .cdg pairs into a single MP3+G track.

[Destination Folder Routing]

1. Target Karaoke Folder:

- Dedicated folder where imported karaoke tracks (.zip, .cdg/.mp3 pairs, and .mp4 video files) are organized.

2. Target Music Folder:

- Dedicated folder where imported background and intermission music (standalone .mp3 files) are organized.

3. Move Files Toggle:

- "Move imported files to destination folders (Karaoke / Music)" automatically moves downloaded files out of Downloads into the designated destination directories upon import.

[Metadata Extraction & Database Indexing]

1. Smart Tagging & Source Identification:

- Analyzes filename, catalog codes (KV / PT / SF), and metadata tags.

- Categorizes track source as "Karaoke Version", "Party Tyme", or "Local".

2. Instant Library Availability:

- Immediately creates or updates the track record in the SQLite database and indexes the song in the SQLite FTS5 search engine.

- Automatically triggers a live library refresh across Karaoke, Rotation, and Playlists views.

[Manual Import & Recent Imports Tracker]

1. "Import Purchased Track..." Button:

- Opens a multi-select file dialog allowing instant manual import of downloaded tracks at any time.

2. Recent Imports Table:

- Real-time audit log showing Source badges, Song Title, Artist, Karaoke Format (MP3G, ZIPCDG, MP4, Audio), File Path, Added Time, and Import Status.

- Includes a "Clear List" button to reset the session view.

Section: Advanced Store Pipeline, Additional Providers & FFmpeg Processing (Lyracist ONLY)

[Additional Search Providers]

In addition to Karaoke Version and Party Tyme, Lyracist supports expanded licensed karaoke catalogs via direct external deep links:

1. "Search Karaoke.com": Direct catalog search for commercial karaoke tracks, MP3+G downloads, and hardware tracks.

2. "Search Sunfly": Deep-link search directly targeting Sunfly Karaoke's professional backing track catalog.

3. Collapsible UI: The "Search Additional Providers" expander keeps the search interface neat while offering instant access to multiple licensed stores.

[FFmpeg Audio & Video Processing Pipeline]

When tracks are imported via the Store watcher or manual import, optional FFmpeg-powered post-processing stages can be activated:

1. Loudness Normalization:

- "Normalize Audio on Import" toggle applies EBU R128 loudness normalization via FFmpeg's loudnorm filter (Target: -16 LUFS, Max Peak: -1.5 dB TP, LRA: 11).

- Eliminates volume jumps between different publishers and track eras.

2. Silence Trimming:

- "Trim Silence on Import" toggle removes leading and trailing silence below -50dB via FFmpeg's silenceremove filter, ensuring immediate music start when playback begins.

3. Preview Waveform Rendering:

- "Generate Waveform Preview" renders a high-contrast PNG visual waveform (showwavespic) for the track, assisting DJs in visual cueing and volume monitoring.

[MP4 Dual-Audio Stream Detection & Lyrics Association]

1. Dual-Audio Stream Detection:

- FFprobe probes MP4 video containers for multiple audio streams (e.g. guide vocal on stream 1, backing music on stream 2).

- Automatically tags tracks as "Dual-Audio" in the library and recent imports table.

2. Automatic Lyric File Pairing:

- Discovers synchronized .lrc and .txt lyric files sharing the same base filename as the audio/video track.

- Automatically pairs the lyric file with the track in the library database.

3. Separate-Arrival MP3+G Reconciliation:

- If a web browser finishes downloading a track's .mp3 and .cdg files a few seconds apart, the watcher buffers and reconciles both files into a single unified MP3+G track.

[Recent Imports & Live Import Event Log]

1. Technical Feature Badges:

- Recent Imports table displays real-time badges indicating active features: Format (MP3G, ZIPCDG, MP4, Audio), Lyrics (paired LRC/TXT), Dual-Audio, Normalization status, and Waveform preview.

2. Import Activity Log:

- Collapsible panel displaying the last 20 import and processing events with timestamps, provider source, status codes (Success, Info, Warning, Error), and processing remarks.

- Includes a "Clear Log" action to reset the audit history.

Section: Lyracist / Karaoke Store & Licensed Track Import / Provider Intelligence (Lyracist ONLY)

[Overview & Purpose]

Provider Intelligence introduces multi-tier forensic fingerprinting into the Lyracist Store import pipeline. When karaoke tracks are imported (via folder watcher or manual import), Lyracist inspects the physical file contents, container structures, and metadata tags to accurately determine the source providerâ€”specifically Karaoke Version, Party Tyme, Sunfly, and Karaoke.comâ€”without relying solely on basic filename patterns.

[Fingerprinting Methodology Hierarchy]

The import pipeline evaluates incoming media through a prioritized 6-tier detection hierarchy:

1. ZIP Internal Signatures (ZIP Inspection):

- Scans central directory and local headers of .zip archives without extracting to disk.

- Karaoke Version: Detected by presence of /custom_backing_track/ directory, exact track.mp3 + track.cdg pair, or "KV" / "Karaoke Version" tokens.

- Party Tyme: Detected by /karaoke/ folder structure, _PT / - PT filename suffixes, or "party tyme" markers.

- Sunfly: Detected by SF prefixed CDG filenames or "sunfly" folder/filename markers.

- Karaoke.com: Detected by KCOM, KARAOKECOM, or karaoke.com filename/folder indicators.

2. CDG Binary Header Fingerprinting:

- Reads the first 24 bytes of standalone or unzipped .cdg graphic stream files.

- Karaoke Version: Signature bytes 0x01 0x0F.

- Party Tyme: Signature bytes 0x02 0x0A.

- Sunfly: Signature bytes 0x03 0x0C.

3. MP3 ID3 Tag Fingerprinting:

- Uses FFprobe to inspect ID3v2 user-defined text frames (TXXX) and comment frames.

- TXXX:KV / TXXX:KaraokeVersion -> Karaoke Version.

- TXXX:PT / TXXX:PartyTyme -> Party Tyme.

- TXXX:SF / TXXX:Sunfly -> Sunfly.

- TXXX:KCOM / TXXX:Karaoke.com -> Karaoke.com.

4. MP4 Container Metadata Fingerprinting:

- Inspects container-level metadata (title, artist, comment, publisher, copyright, and custom tags) via FFprobe.

- Detects brand watermarks and provider names: "Sunfly", "Party Tyme", "Karaoke Version", and "Karaoke.com".

5. Filename Pattern Heuristics (Fallback):

- Evaluates standard naming conventions (e.g. KV*, *Party Tyme*, SF*, KCOM*).

6. Provider Referrer Hint (Fallback):

- Remembers the last provider search button clicked in the Store tab during the current session, attributing downloads to that provider if all internal file signatures are neutral.

Section: Lyracist / Karaoke Store & Licensed Track Import / Smart Import Rules (Lyracist ONLY)

[Overview & Purpose]

Smart Import Rules provide automated post-download curation, file organization, musical classification, and metadata enrichment for imported tracks. Once provider fingerprinting and initial audio probing complete, Lyracist automatically renames files into a canonical standard, classifies genre, estimates vocal performance difficulty, identifies musical key and tempo, detects vocal track configurations, and rates stream quality.

[Auto-Rename Imported Files]

1. Standardized Naming Convention:

- Primary audio/video files and all companion assets are automatically renamed to:

"Artist - Title (Provider).ext"

- Examples:

â€¢ Karaoke Version: "Adele - Hello (KV).mp3"

â€¢ Party Tyme: "Bon Jovi - Wanted Dead or Alive (PT).cdg"

â€¢ Sunfly: "Queen - Don't Stop Me Now (SF).mp4"

â€¢ Karaoke.com: "Frank Sinatra - My Way (KCOM).mp3"

2. Companion File Cohesion:

- Automatically synchronizes base filenames for all paired assets: .cdg graphics, .lrc/.txt lyric files, and .png waveform preview images.

- Eliminates catalog discrepancies and orphaned companion files.

[Auto-Tagging & Technical Classification]

1. Genre Classification:

- Automatically detects and assigns genres based on provider catalog mappings:

â€¢ Karaoke Version: Pop, Rock, Country, Soul, R&B, Jazz

â€¢ Party Tyme: Pop, Rock, Hip-Hop, Gospel

â€¢ Sunfly: Pop, Rock, Dance

â€¢ Karaoke.com: Pop, Rock, Country, R&B, Standards

2. Vocal Performance Difficulty:

- Evaluates performance demand into Easy, Medium, and Hard tiers using tempo (BPM), track length, vocal range spans, and dynamic range.

3. Musical Key Signature Detection:

- Detects standard ID3 frames (TKEY, initialkey) and metadata tags, normalizing keys (e.g. Am, C#m, G) for instant KJ transposer cueing.

4. Tempo (BPM) Detection:

- Identifies and records numeric beats-per-minute (TBPM / tempo) for rotation pacing and DJ transitions.

5. Vocal Presence Classification:

- Identifies audio stream configuration:

â€¢ "guide vocals": Multi-stream dual-audio containers or guide/lead vocal tags.

â€¢ "background vocals": Backing vocal (BGV) and harmony tags.

â€¢ "no vocals": Clean instrumental backing tracks.

6. File Quality Classification:

- Rates media streams into High, Medium, and Low tiers based on audio bitrate (>=256k for High, >=160k for Medium), sample rate, stereo channels, and video resolution (1080p/720p).

[Database & Schema Integration]

- Persisted to SQLite Songs database table (Genre, Difficulty, Key, BPM, VocalPresence, Quality).

- Added to SQLite FTS5 search index tags for instant filtering in the library.

- Reported in real-time within Recent Imports and the Live Import Event Log.

Section: Lyracist / Karaoke Store & Licensed Track Import / Store Analytics (Lyracist ONLY)

[Overview & Purpose]

The Store Analytics panel provides real-time statistical insights and visual dashboards summarizing library composition, provider distribution, file packaging characteristics, FFmpeg processing performance, and temporal import patterns. Integrated directly at the bottom of the Store tab, this collapsible panel empowers host DJs and venue managers to monitor library growth, analyze store acquisition trends, and track audio optimization benchmarks without needing external reporting tools.

[Panel Layout & Controls]

1. Collapsible Section Container:

- Located in the Store tab below the Import Activity Log panel.

- Expand / Collapse Button: Collapses the entire analytics dashboard to save vertical screen space or expands it for detailed inspection. State persists across view navigation.

- Refresh Analytics Button: Instant manual refresh of all LINQ database queries and watcher benchmark metrics.

- Automatic Auto-Refresh: Triggers automatically when the Store tab is navigated to or whenever a new track import event successfully completes.

[Analytics Dashboard Cards]

1. Provider Statistics:

- Total Tracks Imported: Comprehensive count of all cataloged store and library tracks.

- Top Provider Badge: Highlights the most frequently used music provider (e.g. "Karaoke Version (145)").

- Granular Breakdown: Individual track counts and percentage shares for Karaoke Version (KV), Party Tyme (PT), Sunfly (SF), Karaoke.com (KC), and Local/Custom collections.

- Visual Indicators: Color-coded provider markers matching official store branding.

2. File Type & Container Statistics:

- Media Packaging Breakdown: Quantifies track distribution across container formats:

  â€¢ MP3+G Pairs: Discrete audio and graphic file pairs.

  â€¢ MP4 Video: Self-contained MPEG-4 video karaoke tracks.

  â€¢ ZIPCDG Archives: Compressed zip container archives.

  â€¢ Audio-Only (.mp3): Standalone backing tracks and non-karaoke audio files.

  â€¢ Lyrics Files Attached: Counts tracks enriched with external .lrc or .txt synchronized lyric files.

3. FFmpeg Processing & Audio Benchmarks:

- Pipeline Enhancement Metrics:

â€¢ EBU R128 Normalized: Count of tracks loudness-normalized to -16 LUFS broadcast target.

â€¢ Silence-Trimmed: Count of tracks with leading and trailing silence removed below -50dB.

â€¢ Waveform Previews: Count of tracks with visual PNG waveform previews rendered.

- Hardware Performance Benchmarks:

â€¢ Average Processing Time: Mean FFmpeg pipeline execution time per imported file.

â€¢ Longest Processing Time (Max): Peak duration recorded for complex video or normalization jobs.

â€¢ Shortest Processing Time (Min): Fastest turnaround recorded for lightweight audio imports.

4. Quality & Difficulty Distributions:

- Stream Quality Distribution: Categorizes library into Low, Medium, and High fidelity tiers based on audio bitrate (>=256k for High, >=160k for Medium) and video resolution.

- Vocal Performance Difficulty: Categorizes tracks into Easy, Medium, and Hard tiers based on tempo, length, vocal range spans, and dynamic variance.

5. Musical Key & BPM Distribution:

- Top Musical Keys Bar Chart: Visual horizontal bar chart illustrating the top 8 musical key signatures (e.g. C, G, Am, Em) with track counts and relative proportions for rapid KJ transposition planning.

- BPM Tempo Histogram: 5-bucket tempo distribution histogram (<80 BPM, 80-100 BPM, 100-120 BPM, 120-140 BPM, 140+ BPM) showing energy and pacing distribution across the repertoire.

6. Import Activity Timeline & 24-Hour Heatmap:

- Key Activity Summaries: Highlights the busiest calendar date and busiest hour of the day for track acquisitions.

- 14-Day Import Activity Timeline: Daily bar chart displaying acquisition volume over the preceding two weeks.

- 24-Hour Heatmap Grid: 24-hour time block matrix color-shaded by import frequency (Level 0: Dim/Idle, Level 1: 1-2 tracks, Level 2: 3-5 tracks, Level 3: 6-9 tracks, Level 4: 10+ tracks) with interactive tooltips showing exact hourly metrics.

Section: Lyracist / Karaoke Store & Licensed Track Import / Bulk Import Wizard (Lyracist ONLY)

[Overview & Purpose]

The Bulk Import Wizard is a high-throughput batch ingestion pipeline designed for karaoke hosts who purchase track collections in bulk from digital providers or need to import existing libraries into Lyracist in a single streamlined operation. Rather than importing individual tracks one by one, the Bulk Import Wizard allows DJs to select an entire folder, inspect candidate media files, preview extracted metadata and provider origins, configure global and granular FFmpeg audio processing settings, and execute the entire batch with concurrent worker throttling and comprehensive summary reporting.

[Key Features & Operation]

1. Multi-Format Folder Scanning & Audio Pairing:

- Folder Selection: Accessible via the "Bulk Import..." button in Section 4 of the Store tab.

- Supported Media Formats: MP3 audio, CDG graphics, ZIP (MP3+G) archives, MP4 dual-audio video, and LRC/TXT synchronized lyrics files.

- MP3+G Pairing Engine: Automatically detects and pairs corresponding .mp3 audio files and .cdg graphics tracks into a single unified candidate entry, ensuring companion files are never duplicated or treated as standalone orphans.

- Companion Lyrics Linking: Scans adjacent directories and subfolders to automatically match and bind external .lrc or .txt lyric files to candidate tracks.

- MP4 Dual-Audio Stream Detection: Analyzes MP4 audio streams to identify multiplexed lead vocal and instrumental backing tracks.

2. Provider Intelligence Integration & Metadata Preview:

- Automated Origin Detection: Incorporates Lyracist's Provider Intelligence engine to inspect ZIP headers, CDG preamble packets, ID3 metadata tags, and filename patterns. Automatically identifies whether tracks originate from Karaoke Version (KV), Party Tyme (PT), Sunfly (SF), Karaoke.com (KC), or Local collections.

- Comprehensive Candidate Preview Table:

â€¢ Filename: Physical filename of the primary media container.

â€¢ Provider: Detected karaoke provider badge (KV, PT, Sunfly, Karaoke.com, Local).

â€¢ File Type: Container format classification (MP3, MP3+G, ZIP, MP4).

â€¢ Duration: Audio/video duration formatted as mm:ss.

â€¢ Key & BPM: Detected or estimated musical key signature and tempo.

â€¢ Quality & Difficulty: Bitrate tier (Low/Medium/High) and vocal complexity rating (Easy/Medium/Hard).

â€¢ Vocal Presence: Classification of lead vocal tracks (None, Multiplex Left/Right, Backing Vocals).

â€¢ Per-Track Processing Toggles: Granular checkbox overrides for EBU R128 Loudness Normalization, Silence Trimming, and Waveform Generation.

3. Global Batch Options & Targeted File Routing:

- Global Normalization Toggle: "Normalize all" applies EBU R128 loudness normalization (-16 LUFS target) across all candidates.

- Global Silence Trimming: "Trim silence for all" automatically eliminates leading and trailing dead air below -50dB across the entire batch.

- Global Waveform Rendering: "Generate waveform for all" renders visual peak waveform previews (.png) for instantaneous timeline display in the karaoke deck.

- Smart File Routing: "Move files to target folders" routes processed tracks to standardized library destination folders categorized by provider and file type, while applying Smart Import Rules naming ("Artist - Title (Provider).ext").

4. High-Performance Throttled Batch Execution:

- Concurrency Throttling: Employs a multi-threaded execution queue throttled to a maximum of 3 concurrent FFmpeg operations via SemaphoreSlim(3), maximizing multi-core CPU throughput while avoiding disk I/O bottlenecks and keeping the user interface completely fluid.

- Combined Audio Pipeline Pass: When both Silence Trimming and Volume Normalization are enabled, FFmpeg processes both filters in a single combined pass, cutting batch execution time by 50%.

- Real-Time Progress UI: Displays an interactive progress bar, live filename indicator, success counter, failure counter, and an active cancellation token allowing the host to abort remaining jobs gracefully.

5. Bulk Import Completion Summary:

- Completion Modal: Displays a comprehensive post-import summary dialog upon batch finalization:

â€¢ Total tracks imported successfully

â€¢ Total duplicate or existing tracks skipped

â€¢ Total processing errors encountered

â€¢ Unique karaoke providers involved across the batch

â€¢ Average FFmpeg processing time per track and total elapsed duration

- Auto-Refresh Integration: Automatically triggers database synchronization and refreshes Store Analytics dashboards upon window dismissal.

Section: Lyracist / Karaoke Store & Licensed Track Import / Store Sync (1-Click Ingestion) (Lyracist ONLY)

[Overview & Purpose]

Store Sync provides a fast, one-click mechanism for scanning and importing newly downloaded karaoke purchases from digital providers including Karaoke Version, Party Tyme, Sunfly Karaoke, and Karaoke.com. While the background Downloads Watcher operates automatically as new files arrive, Store Sync allows the host to proactively trigger an immediate scan of their configured purchase folder (or system Downloads directory),

inspect and pair incoming media, route them through the Provider Intelligence and FFmpeg audio pipeline, and view a comprehensive post-sync summary dialog.

[Key Features & Operation]

1. 1-Click "Sync Purchased Tracks" Primary Action:

- Dedicated Toolbar Button: Located prominently in Section 4 of the Store tab, styled with an animated sync icon (ArrowSync24) and responsive loading spinner while active.

- On-Demand Trigger: Provides an immediate audit and sync of newly downloaded purchases without requiring application restarts or folder re-selection.

- Flexible Directory Resolution: Automatically targets the custom StorePurchasedTracksFolder configured in Settings, smoothly falling back to the Windows user Downloads folder or standard Music directory.

2. Multi-Format Scanning & Intelligent Filtering:

- Supported File Types: Recursively detects ZIP archives (containing MP3+G or MP4 files), paired MP3 audio and CDG graphics files, standalone MP4 karaoke videos, and accompanying lyrics files (.lrc, .txt).

- Incomplete Download Filtering: Intelligently skips active browser download temp files (.crdownload, .part, .tmp, and ~$ lockfiles) to avoid prematurely processing files while downloads are still in progress.

- Companion Audio & Graphics Pairing: Automatically correlates companion files with identical base names (e.g. "Artist - Title.mp3" and "Artist - Title.cdg"), ensuring they are ingested as a single, cohesive MP3+G track.

3. Automated Pipeline Execution & Smart Deduplication:

- Master Catalog Verification: Checks each candidate track against the SQLite library database. Files already cataloged are safely marked as skipped (TotalSkipped), preventing redundant reprocessing or duplicate entries.

- Provider Fingerprinting: Routes candidate tracks through the Provider Intelligence engine to inspect ZIP archive structure, CDG headers, ID3 metadata tags, and filename patterns to accurately identify provider origins (KV, PT, SF, KC, or Local).

- Smart Import Rules & Target Routing: Applies canonical naming standards ("Artist - Title (Provider).ext"), auto-classifies genre, difficulty, musical key, BPM, and routes tracks into structured library directories.

- FFmpeg Optimization: Executes loudness normalization to EBU R128 standards (-16 LUFS), trims lead-in/trailing dead air silence below -50dB, and generates peak waveform previews.

- Real-Time Audit Trail: Dispatches ImportLogged events to immediately update the Recent Imports tracking grid and audit table.

4. Post-Sync Modal Summary Dialog:

- Detailed Reporting: Upon sync completion, an elegant dark acrylic modal summary window appears presenting:

â€¢ Total Files Scanned: Total count of compatible media files examined in the target directory.

â€¢ Total Tracks Imported: Number of newly cataloged songs successfully added to the library.

â€¢ Total Tracks Skipped: Number of previously existing tracks identified and preserved.

â€¢ Total Errors Encountered: Count of corrupted or unreadable files flagged during processing.

â€¢ Karaoke Providers Involved: Distinct provider tags detected across imported purchases.

â€¢ Average Processing Time & Duration: Per-track processing speed benchmark and total elapsed sync duration.

- Seamless Refresh: Dismissing the summary automatically triggers an instant refresh of Store Analytics dashboards, provider breakdown charts, and audio processing metrics.

Section: Lyracist / Karaoke Store & Licensed Track Import / Track Preview Player (Inspection, Waveforms & Dual-Audio) (Lyracist ONLY)

[Overview & Purpose]

The Track Preview Player provides instant audio-visual inspection and acoustic verification for newly downloaded or cataloged karaoke purchases before or after library import. Integrated seamlessly within the Recent Imports audit grid on the Store tab and the Bulk Import Wizard candidate review list, the preview player allows host DJs to quickly audition tracks, verify audio quality, preview CDG/MP4 video streams, inspect spectral dynamics, and toggle dual-audio vocal channels without launching full karaoke playback sessions.

[Acoustic & Video Preview Engines]

1. 5-Second Rapid Audio Preview:

- Powered by asynchronous FFmpeg stream extraction: extracts the opening 5 seconds of the source track into an uncompressed WAV or MP3 stream stored safely in the system temporary directory.

- Built-in lightweight playback engine utilizes WPF MediaPlayer with accurate millisecond progress tracking, auto-rewind on completion, and manual scrub capability.

- 1-click Play/Pause toggle with real-time elapsed time display (00:00 / 00:05).

2. Dual-Audio Channel Auditioning (Channel A / Channel B):

- Multi-channel stream detection automatically flags dual-audio tracks (such as Party Tyme or Karaoke Version multiplex files featuring lead vocals on one stereo channel and instrumental backing on the other).

- Interactive channel toggle:

• Channel A (Guide Vocals): FFmpeg isolates and routes the left channel (c0) across stereo outputs (pan=mono|c0=c0) for vocal guidance verification.

• Channel B (Instrumental): FFmpeg isolates and routes the right channel (c1) across stereo outputs (pan=mono|c0=c1) for backing track rehearsal.

- Switching channels dynamically re-extracts the 5-second preview and resumes playback seamlessly.

3. Waveform Peak Visualization:

- Generates and renders a high-contrast visual PNG waveform representation using FFmpeg showwavespic.

- Allows hosts to visually inspect audio transients, identify dead air lead-in, and gauge dynamic range at a glance.

4. High-Resolution Spectrogram Visualization:

- Generates a spectral acoustic density map using FFmpeg filter showspectrum=s=800x300:color=intensity.

- Visually reveals frequency distribution across bass, midrange, and treble spectrums, highlighting audio fidelity, clipping, and high-frequency compression artifacts.

5. MP4 Video Stream Preview:

- For MPEG-4 karaoke video files, FFmpeg extracts the opening 5-second video clip (-ss 0 -t 5 -c:v libx264 -preset ultrafast -c:a aac).

- Rendered directly inside an embedded WPF MediaElement with hardware-accelerated playback and looping control.

- Allows immediate visual verification of lyrics animation, title screen graphics, and video resolution.

[Technical Metadata Inspection Dashboard]

The Track Preview Player displays an organized technical badge dashboard highlighting critical musical and audio metadata:

- Filename & Source Provider: Clean canonical naming alongside color-coded provider tag (Karaoke Version, Party Tyme, Sunfly, Karaoke.com, or Local).

- File Type & Container: Format badge indicating MP3G, ZIPCDG, MP4, or Audio.

- Track Duration: Total song playback length formatted as mm:ss.

- Musical Key & BPM: Detected musical key signature (e.g. C, G, Am) and tempo beats-per-minute.

- Stream Quality & Difficulty: Fidelity rating (High, Medium, Low) and performance difficulty tier (Easy, Medium, Hard).

- Vocal Presence: Classification indicating guide vocals, background vocals, or clean instrumental.

[Trigger Points & Integration]

- Store Tab (Recent Imports): Selecting any row in the Recent Imports grid or clicking the dedicated "Preview" button in the Action column instantly displays the Track Preview Player below the table.

- Bulk Import Wizard: Selecting any candidate track in the import review data grid automatically loads its preview, allowing hosts to verify tracks before committing bulk import operations.

Section: Lyracist / Karaoke Store & Licensed Track Import / Provider Settings Panel (Lyracist ONLY)

[Overview & Purpose]

The Provider Settings panel provides centralized configuration of host preferences across the Lyracist Store ecosystem. Integrated directly into the Folder Settings & Processing Pipeline section of the Store tab as a collapsible panel, this interface empowers DJs and KJs to configure their preferred commercial karaoke vendor, preferred media packaging formats, target library routing destinations, baseline FFmpeg audio processing automation flags, and companion lyrics format choices. These preferences seamlessly drive default search behaviors, visual toolbar highlights, Bulk Import Wizard presets, Store Sync ingestion hints, and Smart Import auto-renaming rules.

[Configurable Preferences]

1. Preferred Commercial Provider (KV, PT, Sunfly, Karaoke.com):

  - Configures the default store provider across search actions, UI button highlights, and fallback ingestion hints.

  - Highlights the designated provider button in the Store search toolbar with a gold "★ PREFERRED" badge.

- Pressing Enter inside the Store search box immediately queries the preferred provider's catalog.

- Acts as an intelligent attribution fallback for unbranded files during Store Sync and Bulk Import operations.

2. Preferred File Packaging Type (MP3+G, MP4, Audio-only):

- Establishes the host's preferred karaoke format (MP3+G discrete pairs, MPEG-4 self-contained video, or standalone Audio backing tracks).

- In Bulk Import Wizard: Automatically prioritizes and sorts candidate tracks matching the preferred packaging format to the top of the review grid.

- In Smart Import Rules: Serves as a classification hint when disambiguating companion files.

3. Baseline Audio Processing Automation Flags:

- "Normalize audio by default": Automatically pre-checks EBU R128 loudness normalization (-16 LUFS) for folder watcher imports and Bulk Import Wizard operations.

- "Trim silence by default": Automatically pre-checks leading/trailing dead air silence removal below -50dB.

- "Generate waveform by default": Automatically enables high-contrast visual PNG waveform preview rendering.

4. Preferred Target Folder Routing (Karaoke or Music):

- Sets the default destination directory category when auto-routing downloaded purchases into organized library folders.

- Directs karaoke containers (.zip, .cdg/.mp3, .mp4) to the Target Karaoke Folder or standalone audio files to the Target Music Folder.

5. Preferred Synchronized Lyrics Format (LRC or TXT):

- Configures priority format resolution when multiple companion lyric files exist in the same download directory.

- When set to "LRC", the scanner searches for and pairs .lrc synchronized files first, falling back to plain .txt.

- When set to "TXT", plain text files take precedence over .lrc.

[User Interface & Controls]

- Dropdown: Preferred Provider (Karaoke Version [KV], Party Tyme [PT], Sunfly [SF], Karaoke.com)

- Dropdown: Preferred File Type (MP3+G, MP4, Audio-only)

- Dropdown: Preferred Target Folder (Karaoke, Music)

- Dropdown: Preferred Lyrics Format (LRC, TXT)

- ToggleSwitch: Normalize audio by default

- ToggleSwitch: Trim silence by default

- ToggleSwitch: Generate waveform by default

- Action Buttons: "Save Settings" (persists to app configuration with instant visual confirmation) and "Reset Defaults" (reverts to standard factory defaults).

Section: Lyracist / Karaoke Store & Licensed Track Import / Store Plugin API (Lyracist ONLY)

[Overview & Purpose]

The Store Plugin API establishes a modular, extensible plugin architecture for the Lyracist Store ecosystem. Commercial karaoke providers (Karaoke Version, Party Tyme, Sunfly, Karaoke.com, and future additions) are encapsulated as independent, self-contained plugins adhering to the IStoreProvider interface. This architecture centralizes search URI generation, filename heuristics, ZIP archive inspection, CDG binary header matching, ID3 tag identification, and MP4 container metadata analysis into dedicated provider modules managed by a thread-safe ProviderRegistry.

[Architecture & Core Components]

1. IStoreProvider Interface:

- string Name { get; }: Display title of the commercial provider (e.g., "Karaoke Version").

- ProviderSource Source { get; }: Provider origin enum value (KaraokeVersion, PartyTyme, Sunfly, KaraokeCom, Local).

- Uri BuildSearchUri(string query): Formats deep search URLs for catalog queries or homepages.

- bool DetectFromFilename(string filename): Evaluates filename and title/artist heuristic patterns.

- bool DetectFromId3(FFprobeResult metadata): Evaluates ID3 audio tags (e.g. TXXX frames).

- bool DetectFromZip(string zipPath): Scans internal archive hierarchy and file pair layouts.

- bool DetectFromCdgHeader(byte[] header): Matches initial CDG magic header bytes.

- bool DetectFromMp4(FFprobeResult metadata): Analyzes MP4 stream metadata and container tags.

2. BaseStoreProvider Abstract Base Class:

- Provides default false implementations for all detection methods, allowing plugins to override only their supported signatures.

- Encapsulates safe ZIP entry enumeration (MatchesZipEntries), ID3 tag scanning (MatchesId3Tags), and MP4 metadata inspection (MatchesMp4Metadata).

3. Concrete Store Providers:

- KaraokeVersionProvider: Search URL for karaoke-version.com, detection of /custom_backing_track/ or track.mp3+track.cdg pairs, KV prefix/tags, and CDG magic header bytes 0x01 0x0F.

- PartyTymeProvider: Search URL for partytyme.net, detection of /karaoke/ folders, _pt. naming, PT tags, and CDG magic header bytes 0x02 0x0A.

- SunflyProvider: Search URL for sunflykaraoke.com, detection of sf entry prefixes, SF tags, and CDG magic header bytes 0x03 0x0C.

- KaraokeComProvider: Search URL for karaoke.com, detection of kcom signatures, and ID3/MP4 metadata watermarks.

4. ProviderRegistry Central Registry:

- Maintains a thread-safe catalog of registered provider plugins.

- Supports dynamic registration via RegisterProvider(IStoreProvider).

- Provides unified multi-vector discovery APIs: DetectProviderFromFilename, DetectProviderFromZip, DetectProviderFromCdg, DetectProviderFromId3, and DetectProviderFromMp4.

- Directly drives background download watcher ingestion, Bulk Import Wizard classification, Store Sync verification, and StoreViewModel search button generation.

Section: Lyracist / Karaoke Store & Licensed Track Import / Store Notifications (Lyracist ONLY)

[Overview & Purpose]

The Store Notifications system provides non-intrusive, animated toast alerts that inform users in real time as tracks are processed and cataloged into their Lyracist library. Designed to eliminate guesswork during background downloading and batch ingestion, notifications appear in a dedicated bottom-right overlay container inside the Store tab and the Bulk Import Wizard. Each toast features smooth slide-in/fade-in entry animations, category iconography, detailed track attribution, processing badges, a manual dismiss button, and an automatic 5-second dismissal timer.

[Notification Triggers & Types]

1. Track Imported:

- Alert: "Imported: Artist - Title (Provider)"

- Badges: Packaging format (MP3+G, MP4, ZIP) along with pipeline flags (NORM, TRIM, WAVE).

- Trigger: Dispatched immediately when a purchased track finishes metadata probing, renaming, audio processing, and library database indexing.

2. Audio Normalization Complete:

- Alert: "Audio normalized: Artist - Title"

- Trigger: Dispatched when EBU R128 loudness normalization (-16 LUFS) completes successfully.

3. Silence Trimmed:

- Alert: "Silence trimmed: Artist - Title"

- Trigger: Dispatched when leading and trailing dead-air silence below -50dB is removed via FFmpeg.

4. Waveform Generated:

- Alert: "Waveform ready: Artist - Title"

- Trigger: Dispatched when the high-contrast acoustic waveform preview image finishes rendering.

5. Store Sync Completed:

- Alert: "Store Sync Complete â€” X tracks imported"

- Trigger: Dispatched when the 1-Click Store Sync scanner finishes evaluating the download directory.

6. Bulk Import Completed:

- Alert: "Bulk Import Complete â€” X tracks processed"

- Trigger: Dispatched when a batch import job completes inside the Bulk Import Wizard.

[UI Controls & Architecture]

- ToastNotificationControl: High-DPI dark acrylic card (#23273A) with subtle border (#384260), drop shadow, category accent badge, wrap-around message text, badge strip, and manual close button.

- NotificationHost Overlay: Positioned at the bottom-right corner of StorePage.xaml and BulkImportWindow.xaml, stacking up to 5 active toasts vertically without obstructing user interactions or scrolling page content.

- StoreNotificationService: Central thread-safe notification manager providing event dispatching, thread marshalling to the WPF UI Dispatcher, and auto-dismiss timer cancellation.

Section: Integration Settings / Party Tyme Removal

[Update Details]
The Party Tyme karaoke streaming integration has been removed from the application.
1. Removal of Streaming Tab:
   - The "Party Tyme Online" tab has been removed from the main Karaoke dashboard view.
   - The "Party Tyme (Streaming)" tab has been removed from the Special Occasions Music search dialog.
2. Settings Panel Clean-up:
   - The subscription fields (Client ID and Client Secret) for Party Tyme have been removed from the Settings panel under APIs & I
3. Queue & Badge Updates:
   - The "Party Tyme" filter checkbox and visual badge triggers have been removed from the singer rotation queue list.

Section: Display Projection Window / Casting Support

[Update Details]
Operators can now select different casting targets for the Singer Rotation Billboard screen in both Lyracist and KSRotation applica
1. Casting Target Dropdown:
   - Located in the "Settings" tab under "Displays & Projection" (or "Display & Projection" in KSRotation).
   - Dropdown options include Monitor, Miracast, Chromecast, Browser Cast, AirPlay, and Wireless HDMI.
   - When a casting target is selected, the application (Lyracist or KSRotation) initializes the appropriate casting sender/server in th
2. Off-Screen Rendering:
   - Selecting a casting target (such as Chromecast or Browser Cast) creates and renders the Rotation Window / Singer Display W
   - This ensures the KJ laptop screen is not cluttered while casting frames.
3. Browser Cast Server:
   - The self-hosted Browser Cast option spins up a local web server (http://localhost:8080/rotation/) to allow any browser on the lo

Section: Database Manager / Catalog Book Exporter & Failed Artist List

[Update Details]

Updates have been made to the Catalog Book Exporter formats and the Slow Metadata Extractor.

1. Supported Formats and Sorting:
   - The Catalog Book Exporter now supports exporting the catalog as a plain text (.txt) file.
   - The format selection dropdown has been sorted in the following order: PDF Document (.pdf), Text File (.txt), and Word Docum
2. Metadata Scan Failed Artist Export:
   - When running a Slow Metadata Scan for songs with unknown artists, any files that were unable to be resolved and updated wi
   - Upon completion or cancellation of the scan, a plain text file listing all unresolved file paths is automatically exported to the "Re

Section: Database Manager / Catalog Book Exporter

[Update Details]
Operators can now export the entire song database (Karaoke or Music) into Word (.docx) or PDF format directly from the Lyracist

1. Exporter Panel:
  - Located in the right pane of the Lyracist Database Manager under "Catalog Book Generator".
  - Provides two dropdown selectors:
    - Catalog: Choose between the "Karaoke Catalog" (tracks marked as karaoke) and the "Music Catalog" (standard audio tracks)
    - Format: Choose between "Word Document (.docx)" and "PDF Document (.pdf)".

2. Generated Fields & Formatting:
  - The generated books are structured as clean, professionally styled tables showing:
    - Artist: Alphabetically ordered performer names.
    - Title: Song titles.
    - Length: Formatted track durations (MM:SS).
    - Genre: The embedded audio genre (automatically populated during metadata scans).

3. Pagination & Output:
  - The document is fully paginated, displaying "Page X" or "Page X of Y" in the bottom footer.
  - For Word (.docx) books, a repeating header row is defined so column titles automatically reappear on the top of each page.
  - When generation completes, Lyracist automatically launches the default system viewer to open and print the document, which

Section: Performer Profiles, Song Tagging & Singer Avatars

[Update Details]
A comprehensive performer profile system and song tagging support have been integrated into Lyracist.
1. Performer Profiles & Login:
   - Patrons accessing the Patron Request Portal can log in or register using their Performer Name and a 4-digit PIN.
   - Profile settings include email registration, custom vocal range specification (e.g. Tenor, Soprano), and custom title assignment
2. Singer Avatars & Custom Uploads:
   - Performers can customize their profile avatar using a dropdown selector.
   - Supports default silhouette, Gravatar logo generation (via email MD5 hash), and direct mobile photo/selfie uploads (capped at
   - Avatars are displayed as circular icons in the rotation board on both the mobile Patron Portal and host WPF Singer Queue List
3. Database Tagging:
   - Songs can now be tagged with comma-separated categories to organize, filter, and challenge performers.

Section: Karaoke Song Search & Compact Auto-Advance Toolbar

[Update Details]

Optimizations have been introduced to the primary Karaoke hosting workspace for database search and Auto-Advance DJ control

1. Direct Song Search Database Scan:
   - Clicking the 'Scan' button beside the search input box immediately scans and filters songs from the local SQLite database mat
   - Pressing the Enter key while typing inside the search input box also immediately executes the database scan.
   - Disk folder scanning is cleanly separated into an adjacent folder icon button ('Scan music folder from disk into library...'), preve
2. Compact Auto-Advance DJ Toolbar:
   - The Auto-Advance control strip has been redesigned into a slim, space-efficient horizontal toolbar (reducing vertical height fron
   - DJ controls ('Start Song' and 'Skip Singer') have been streamlined into compact action buttons with informative tooltips.
   - This reclaims significant vertical screen real estate for the song results table, singer queue, and live lyrics preview panels on la

Section: Audience Lyrics Screen QR Code & Library Format Separation

[Update Details]

Major updates have been added to improve projection screen customization and local library management across karaoke and sta

1. Audience Lyrics Screen QR Code Toggle:
   - Hosts can now toggle the display of the patron mobile request QR code badge positioned in the top-right corner of the lyrics pr
   - Configurable in three intuitive locations:
     - Settings -> Monitors & Screen Assignments: 'Show QR Code on Lyrics Screen' checkbox under Lyrics Projection Screen.
     - Lyrics Page: 'Show QR Code on Lyrics Screen' toggle switch under the Audience Lyrics Screen settings card.
     - Lyrics Window Context Menu: Right-click anywhere on the active lyrics projection window and click 'Show QR Code' to toggle
   - The toggle state is permanently persisted across application restarts in AppSettings.
2. Karaoke Library vs. Music Library Separation:
   - Automatic MP3+G Companion CDG Detection: The library scanner now automatically identifies unzipped karaoke pairs (e.g. s
   - Expanded Audio Format Support: The library scanner now supports comprehensive audio file types including .mp3, .mp4, .zip,
   - Distinct Library Tabs with Live Count Badges: The search results view now features 'Karaoke Library' and 'Music Library' tabs
   - Search Query Isolation: SQLite database queries now evaluate track type filters directly in SQL, ensuring music searches retur

Section: Search Performance, Tablet Touch Sizing & High-Output Audio Engine

[Update Details]

Major performance, layout responsiveness, and audio output enhancements have been integrated into Lyracist for seamless live D

1. High-Performance Instant Search & Lookup:
   - Zero-Lag Search Results: Song search results now batch UI notifications, updating the Karaoke and Music library tables instan
   - Isolated Streaming Queries: External network searches (YouTube, Spotify, Amazon) are isolated exclusively to the Streaming
   - Quick-Clear Search Button: A dedicated clear button ('X') has been integrated directly into the search bar to clear queries and
   - Active Row Virtualization: Search grids now employ full UI recycling virtualization, ensuring instant scrolling even with large cat
2. Responsive Tablet Touch Sizing & Column Protection:
   - Tablet-Optimized Action Buttons: Action buttons in search result tables (Add to Rotation, Singer History, and Singer Queue) ar
   - Anti-Squish Protection on High-DPI Displays: On 1080p tablet screens operating under 150% or 175% Windows DPI scaling, a
   - Minimum Panel Width: The search and assignment panel enforces a 360px minimum width to ensure readable song titles and
3. Punchy High-Output Audio Playback:
   - Native Windows WASAPI mmdevice Audio Pipeline: All karaoke playback (LibVlcVideoBackend) and background music chann
   - Equalizer Preamp Stabilization: Hardware and software equalizer filters now initialize with clean 0 dB preamp gain, eliminating
   - Enhanced DJ Volume Headroom: Software volume and gain headroom have been expanded up to 150% (+6 dB digital boost),
   - Calibrated Live Loudness Target: Automatic volume normalization has been recalibrated to -12 LUFS (matching commercial liv

Section: Master Output Boost / USB Mixer Mode & Anti-Clipping Limiter

- Master Output Boost / Preamp (USB Mixer Mode): Provides an adjustable digital preamplification slider from 0 dB to +12 dB with
- Optimized for Professional USB Mixers: Engineered specifically for external sound cards and USB audio interfaces like the Yama
- Anti-Clipping Peak Limiter: Integrated transparent soft-knee peak compressor/limiter that safeguards against digital clipping and
- Expanded Volume Headroom: Playback volume headroom ceiling extended to 200% (+6 dB digital gain) across both the main ka
- Unified BGM and Performance Levels: Master preamp boost settings automatically apply across all playback channels (Karaoke

Section: Karaoke Screen 1080p Resolution Layout Optimization

- 1920x1080 Resolution Optimization: Refined page margins and layout heights to eliminate vertical clipping on standard Full HD (
- Compact TitleBlock & Reclaimed Real Estate: Replaced the large 100px+ header with a sleek 40px toolbar featuring inline Venu
- Interactive QR Code Thumbnail & Quick Popout: Scaled the inline QR code to a clean 30x30 thumbnail with Join URL and 1-clic
- Internal Search Results DataGrid Scrolling: Removed the outer column ScrollViewer so the Karaoke and Music Library DataGrid
- Direct Singer Assignment Visibility: Redesigned the Singer Assignment card into a compact two-column layout (Singer Name + D

Section: Special Event Banner Single Selection & 'None' Button Reset

- Guaranteed Mutual Exclusion: Special Event Banners (e.g. Birthday, Anniversary, Holiday, Scaryoke) now strictly enforce single-
- Active 'None' Selection Reset: Selecting the 'None' radio button cleanly clears and deselects all active special event banner sele
- Synchronization & Persistence: The active selection is synchronized seamlessly between Lyracist and KSRotation through the e

Section: Hybrid GPS Venue Auto-Location & DashCast TV Streaming

- Hybrid GPS & Wi-Fi Venue Detection: Automatically recognizes performance venues using a 150-meter GPS proximity circle (via
- Travel Router SSID Immunity: Portable travel routers broadcasting static SSIDs across multiple gigs can be marked as 'Travel R
- Tablet-to-Laptop GPS Sync: Laptops without satellite GPS hardware receive real-time peer GPS coordinates from companion An
- 1-Click '□ Tag GPS' Geotagging: Instantly records and saves current satellite coordinates, Wi-Fi SSID, and 150m detection radi
- DashCast TV Billboard Streaming: Stream live rotation queues, current singer, upcoming queue, and dual QR codes directly to C

Section: Singer Skip Round-Scoped Rotation Bypass

- Singer Skip vs. Inactive vs. Paused: When a performer needs to step away temporarily (e.g. grabbing a drink or stepping outside
- Retains Rotation Placement: Unlike 'Inactive' (which forfeits rotation spot and drops the singer to the end of the line), a skipped s
- Automatic Round-Scoped Rollover: When rotation advances and the round completes (crossing or reaching the '□ Anchor' singe
- Manual Unskip: The DJ can toggle 'Skip' off at any time using the row button, context menu, or remote DJ portal. If the currently
- Cross-Platform UI & Indicators: Supported with visual '□ SKIPPED' amber badges, quick action toggle buttons, and synchronize

Section: Special Singer (One-Time Performance)

- Special / Guest Performer Support: Accommodates guest singers or one-off performances who only perform a single song witho
- Visual Highlight While Performing: Displays a prominent '□ SPECIAL' badge on stage, detached rotation displays, external audi
- Automatic Inactive Transition: Immediately following the conclusion of their song, the special performer is marked inactive autom
- Seamless Queue Resumption: The rotation immediately resumes with the displaced performer or next sequential singer as if the
- Rotation Anchor Immunity: Round anchor allocation algorithms ignore special performers so round tracking boundaries are neve

Section: Special Singer (One-Time Performance)

- Special / Guest Performer Support: Accommodates guest singers or one-off performances who only perform a single song witho
- Instant Top of List & Current Singer: When added or designated, the special singer is placed immediately at the top of the queue
- Displaced Performer Continuity: If another performer was already singing or designated next, they are preserved as 'Up Next' so
- Visual Highlight While Performing: Displays a prominent '□ SPECIAL' badge on stage, detached rotation displays, external audi
- Automatic Inactive Transition: Immediately following the conclusion of their song, the special performer is marked inactive autom
- Seamless Queue Resumption: The rotation immediately resumes with the displaced performer or next sequential singer as if the
- Rotation Anchor Immunity: Round anchor allocation algorithms ignore special performers so round tracking boundaries are neve

Section: Seamless Bidirectional Device Switching & Session Handoff

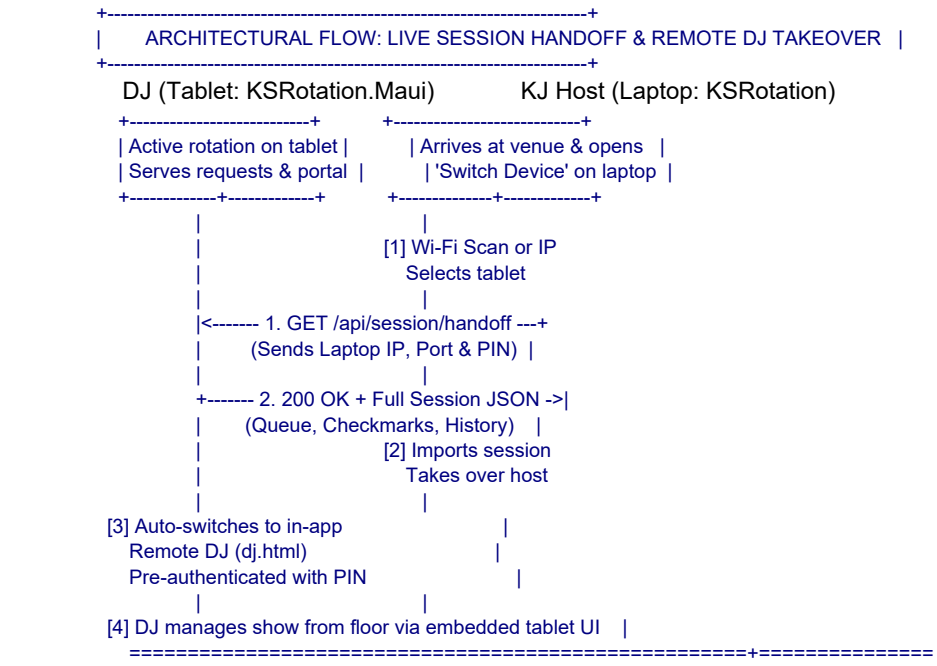
- Live Karaoke Session Continuity: Allows a host or DJ to transition an ongoing karaoke event between a laptop (running KSRotat
- Complete State Preservation: The handoff payload completely transfers the active singer queue, currently performing singer ('Is
- Dedicated Switch Device QR Code & Web Portal: The hosting device generates a high-resolution QR code pointing to http://<ho
- Native Deep Linking (ksrotation://handoff): When scanned or tapped on mobile/tablet, the system deep-links directly into KSRota
- One-Tap Wi-Fi Auto-Discovery: Both KSRotation and KSRotation.Maui feature an inline 'Scan Wi-Fi' network discovery tool that
- Bidirectional Handoff (Laptop <-> Tablet): Supports transferring from Laptop to Tablet when leaving the venue, and transferring f
- Failsafe Manual Entry & Session Backup (.ksr): If local router client isolation is enabled, DJs can enter the IP and DJ PIN manua

Section: Automated Transition to In-App Remote DJ Controller on Handoff

- Automatic Tablet Remote DJ Transition: When the host arrives and pulls an active karaoke session from the DJ's tablet using KS
- Complete In-App Experience: The DJ remains inside the native KSRotation.Maui application, where an embedded full-screen Re
- Automatic PIN Pre-Authentication: The desktop host securely passes its connection address, port, and DJ PIN during the handof
- Top Header Controls & Standalone Exit: The in-app Remote DJ view features a dedicated header toolbar with live connection inc
- Configurable Auto-Switch Setting: An 'Auto-switch to Remote DJ (dj.html) on handoff' toggle is available directly on the Switch De
- Manual Remote DJ Connection: DJs can also tap the 'Connect as Remote DJ' button directly from the Switch Device dialog to co

Section: Device Switching, Live Session Handoff & Architecture Flow

[Architectural Flow: Live Session Handoff & Remote DJ Takeover]



Seamless bi-directional synchronization over venue Wi-Fi!

[Key Capabilities & Workflow]

1. Zero-Beat Live Session Migration:
   - Completely transfers active queue, current performer, up-next performer, rotation start anchor, round checkmarks (1-10), tonight's performanc
2. Automated In-App Remote DJ Transition:
   - When the host laptop takes over, the DJ's tablet automatically transitions into an embedded full-screen Remote DJ controller (dj.html), pre-aut
3. One-Tap Wi-Fi Auto-Discovery:
   - Sweeps the venue subnet in 1 second to detect active instances on port 5000. Selecting a discovered peer auto-populates the host IP and por
4. Dedicated Switch Device QR Code & Deep Linking:
   - Generating a QR code to /handoff allows instant camera scanning, opening a mobile landing page with a 1-tap 'Open in KSRotation MAUI' dee
5. Dedicated Kiosk Request Station (kiosk.html):
   - Mount an Android or iOS tablet in landscape mode at the bar or near the stage. When idle, an attractive attractor screen greets patrons, invitin
6. Audience Billboard & Chromecast Web-Casting (billboard.html):
   - Stream the live audience billboard directly to Google Cast / Chromecast displays from KSRotation.Maui, featuring a dedicated full-height rotati

Section: Dynamic Round Completion Estimation & Full Round Duration Notice

[Overview & DJ Benefits]

• End-of-Night Round Timing:
  - During live karaoke shows, knowing whether there is sufficient venue time remaining for another full round before closing or last call is
  - The Dynamic Round Completion Estimation system tracks the remaining performers in the active round, calculates remaining minutes

[Cross-App Availability & Displays]

1. Remote DJ Controller (dj.html):
   - A prominent, color-accented status banner is docked directly beneath the Last Round banner at the top of the queue panel.
   - Shows real-time performer count left in the round, remaining minutes, projected clock completion time (e.g. 'ends ~11:42 PM'), and full
   - Updates dynamically every 15 seconds, and recalculates immediately on queue modifications (singer add, reorder, status toggle).
2. Desktop Karaoke Station (KSRotation):
   - Styled badge beside active singer counts in the main Rotation header toolbar: '□ □ Round: 4 singers left • ~20m (ends ~11:42 PM) | F
3. Mobile Tablet App (KSRotation.Maui):
   - Pinned estimation banner above the singer rotation list on Android and Windows tablets.
4. Lyracist Pro (Rotation Page):
   - Pinned estimation banner above the singer queue on the Rotation management page.

[Smart Round Mechanics]

• Dynamic Recalculation:
  - Updates as performers finish songs, new singers are added, or songs are reordered.
• Last Round Awareness:
  - In 'Last Round' mode, filters strictly for performers who have not yet sung in the final round ('!HasSungInLastRound'), showing the exac
• Singer Status Filtering:
  - Automatically excludes paused, skipped, inactive performers, and filler background music tracks so timing projections reflect true active

Section: Active Singers Count Display on Vegas Billboard and Vinyl Record Banners

[Overview & Audience Display Enhancements]

• Live Rotation Count Visibility:
  - Both the Vegas Marquee ('Vegas Billboard') and the Vinyl Turntable ('Vinyl Record') projection banners prominently display the live cou
  - The badge reflects real-time rotation state, instantly updating as singers are added, reordered, marked inactive, or when a final round is

[Display Locations & Styling]

1. Vegas Marquee ('Vegas Billboard'):
   - Top Header: A sleek, gold-bordered glowing badge ('□ {N} Singers in Rotation') is anchored in the top-right corner of the brass marqu
   - Up Next Section: A matching pill badge sits directly alongside the 'UP NEXT' label above the upcoming singer chips.
2. Vinyl Turntable ('Vinyl Record'):
   - Now Spinning Header: A glowing gold badge ('□ {N} Singers in Rotation') is positioned alongside the '♪ NOW SPINNING' / '♪ NOW PL
   - On Deck Section: A matching pill badge sits directly alongside the 'ON DECK' queue label.

[Smart Filtering]

• Pure Vocalist Tracking:
  - Excludes paused, skipped, and inactive singers, as well as background music filler tracks, so patrons and the KJ see the exact count o

Section: Four New Dynamic Rotation Billboard Projection Views

[Overview & Audience Display Options]

Lyracist Pro and KSRotation now feature eleven distinct, high-impact projection themes for the audience-facing Singer Rotation Billboard screen.

1. Casino Slot Reels:
   - Experience high-roller casino excitement with each rotation slot rendered as a spinning slot machine reel.
   - When a performance starts, the hero performer reel rapidly spins and clunks to a decelerated stop on the payline.
   - The current singer receives a celebratory "JACKPOT" flourish complete with a pulsating golden drop-shadow glow and a dynamic burst of drif
2. Jukebox:
   - Transports the venue to a classic 1950s rock-and-roll diner cabinet.
   - Features a chrome-trimmed neon arch header, glowing illuminated song-selection pushbuttons for the upcoming queue, and dual animated ris
3. Stadium Jumbotron:
   - Delivers a massive arena sports-and-concert experience with an authentic LED scoreboard aesthetic.
   - Performer names and song titles are presented with high-visibility dot-matrix typography while twin sweeping stadium floodlights crisscross the
4. Movie Theater 'Now Showing':
   - An elegant vintage Hollywood premiere presentation.
   - Begins with a classic 35mm film leader countdown sweep (3...2...1) and a flickering cinematic projector light cone that cascades over the featu
   - Upcoming singers are showcased on vintage marquee coming-attractions lobby cards.

[Full Feature Parity & Controls]

• Complete Badge Support:
  - All four new views fully support performer avatars/selfies, □ SPECIAL performer badges, □ ANCHOR round-start markers, and {N} estimated
• Automatic Cycling & Rotation Schedule:
  - All eleven projection views are dynamically integrated into the Screen Rotation schedule under Settings -> Display -> Screen Rotation.
  - DJs can check which views to include and configure custom display durations (in seconds) for hands-free, automated cycling throughout the sh
• View Mode Help & Descriptions:
  - Complete descriptions for all eleven themes are accessible directly within KSRotation (Help Topic 3) and Lyracist Pro (Settings -> Display -> B

Section: Screen Rotation and Display Tab Layout Reorganization

[Overview & Layout Redesign]

The Display configuration experience in KSRotation and Lyracist Pro has been streamlined for enhanced operational clarity and fast setup:

1. Two-Column Optimized Layout:
   - Target Monitor (Display & Projection), Connect & Request Instructions, and Casting Controls are consolidated into Column 0 on the left.
   - The entire Column 1 on the right is dedicated to the Screen Rotation settings, providing a clean, uncluttered interface.
2. Unified Rotation Interval (Single Time Setting):
   - Eliminated tedious individual per-screen duration inputs in favor of a single global duration setting (e.g., 'Change screen every: 180 sec').
   - DJs and KJs no longer have to manually set the seconds on every individual view.
3. Random Screen Cycling:
   - When automatic screen rotation is active, the billboard projection now randomly cycles among the enabled views at every configured interval.
   - When multiple screens are enabled, the random picker automatically avoids immediately repeating the current screen, delivering dynamic and
4. One-Click 'Select All' & 'Clear All' Controls:
   - Added instant '□ Select All' and '□ Clear All' action buttons above the screen list.
   - Hosts can enable all 11 billboard views with a single click or clear the selection instantly to focus on just their favorites.

Section: Song End Return to Karaoke Screen & DJ Manual Start Workflow

[Overview & Operational Flow]

To ensure smooth hosting transitions during live shows, Lyracist Pro provides an optimized workflow when a singer completes their performance:

1. Automatic Return to Karaoke Control Screen:
   - When the currently playing song finishes, the main host window immediately navigates back to the primary Karaoke screen (KaraokePage).
   - This provides the DJ with instant access to the singer queue, search bar, and playback controls without requiring manual tab clicks.
2. Audience Display Reset to Rotation Billboard:
   - The secondary projection window seamlessly closes the lyrics view and restores the full-screen Rotation Billboard.
   - The billboard displays the upcoming singer queue and automatically flashes an announcement welcoming the next singer to the stage.
3. DJ-Controlled Performance Start (No Automatic Song Progression):
   - Rather than automatically launching the next song over an empty stage, Lyracist marks the completed singer done, queues the next performe
   - The DJ initiates playback with the '□ Start Song' button once the next singer is stationed on stage with their microphone.

Section: Lyrics Screen Audio Visualizer & Spectrum Customization

[Overview & Operational Capabilities]

Lyracist Pro includes a configurable, real-time audio spectrum visualizer displayed across the bottom of the active Lyrics projection screen:

1. Live Audio Spectrum Analysis (Real-Time WASAPI Loopback FFT):
   - Rather than simple simulated animation, Lyracist features a hardware-accelerated Windows Core Audio (WASAPI mmdevice) loopback engin
   - A 1024-point Radix-2 Fast Fourier Transform (FFT) with Hann windowing decomposes music into logarithmic frequency bands (from 32 Hz su
   - Frequency pre-emphasis tilt ensures high-frequency percussive elements (hi-hats, cymbals) dance vigorously alongside bass kicks and vocal
2. Simulated & Ambient Motion Mode:
   - For hosts preferring purely decorative motion without audio loopback capture, hosts can switch to 'Simulated / Ambient' mode which animates
3. Visual Themes & Color Schemes:
   - Choose between 7 stylized gradient themes:
     * Neon Sunset: Blue (#2563EB) to Purple (#A855F7) to Hot Pink (#EC4899).
     * Cyberpunk: Electric Cyan (#00F0FF) to Deep Violet (#7928CA) to Neon Fuchsia (#FF007F).
     * Emerald Pulse: Forest Green (#065F46) to Vibrant Jade (#10B981) to Neon Lime (#84CC16).
     * Solar Flare: Crimson Red (#DC2626) to Vivid Amber (#F59E0B) to Electric Gold (#FDE047).
     * Electric Blue: Deep Navy (#1E3A8A) to Azure (#0284C7) to Ice Cyan (#67E8F9).
     * Rainbow Spectrum: 360-degree full-spectrum rainbow hue mapping across every individual frequency bar.
     * Monochrome Glow: Deep Slate (#334155) to Silver (#94A3B8) to Crisp White (#FFFFFF).
4. Bar Width, Density & Spacing Adjustments:
   - Slim: 10px bars (high density, up to 80 bars).
   - Normal: 18px bars (standard balanced presentation, ~40 bars).
   - Wide: 28px bars (bold, chunky retro aesthetic, ~25 bars).
   - Extra Wide: 42px bars (classic broadcast DJ equalizer look, ~16 bars).
5. Opacity Control & Instant Context Menu Access:
   - Adjust visualizer opacity from 10% to 100% via Settings or quick-select (25%, 50%, 80%, 100%).
   - All visualizer settings (Toggle, Mode, Style, Bar Width, Opacity) can be adjusted in real-time from the Settings Page (Display Tab) or by right-c

Section: DJ Karaoke Screen / Adjust Synth Display Pop-up Dialog

[Update Details]

To ensure a completely professional live performance experience without audience distraction, DJs and KJs no longer need to interact with or rig

1. Dedicated 'Adjust Synth Display' DJ Controls:

- Placed directly on the primary 'Karaoke' operational screen in multiple convenient locations:
  * Next to the 'Skip Singer' button on the top DJ playback control strip.
  * Above the live 'Lyrics Preview Monitor' in the right-hand inspection column.
  * Alongside the 'Lyrics Projection Screen' monitor assignment controls under Multi-Monitor Display controls.

2. Modal Settings Pop-up Dialog ('Adjust Synth Display'):

- Clicking 'Adjust Synth Display' opens a dedicated modal dialog centered over the main Lyracist DJ window (completely invisible to the audienc
- Allows instant real-time adjustment of:
  * Visualizer Enable/Disable toggle.
  * Motion Mode: 'Audio Spectrum (Live FFT)' (reactive loopback frequency bands) vs 'Simulated / Ambient' (procedural sine waves).
  * Color Themes: Neon Sunset, Cyberpunk, Emerald Pulse, Solar Flare, Electric Blue, Rainbow Spectrum, Monochrome Glow.
  * Bar Width & Density: Slim, Normal, Wide, Extra Wide.
  * Opacity Control: Continuous slider (10% to 100%) plus 25%, 50%, 80%, and 100% quick preset buttons.
  * Real-time Interactive Preview: A live animated preview strip inside the dialog provides immediate visual feedback of color, density, and opaci

3. Persistent Configuration:

- All visualizer adjustments made in the dialog apply to the active projection screen immediately and are automatically saved to appsettings.jso

Section: Audience Welcome & Pre-Show Screen / Dual QR Code Integration

[Update Details]

To optimize patron onboarding before a show begins and between active performance sets, the venue projection display now features a dedicate

1. Pre-Show Screen Mode ('Hold Screen'):

- A dedicated 'Pre-Show Screen' toggle button is accessible directly on the primary DJ control toolbar in both Lyracist and KSRotation, as well a
- When engaged, the Audience Sign-Up / Welcome screen remains actively displayed on the singer/audience projection monitor even after sing
- This allows DJs and KJs to set up and populate their queue ahead of time while keeping the audience welcome, Wi-Fi credentials, and sign-up
- A configurable hotkey shortcut (customizable in Settings > Hotkeys) is provided to toggle Pre-Show Mode instantly from anywhere in the appli

2. Stacked QR Code Layout (~20% Screen Width):

- The right side of the 1080p audience canvas is dedicated to clear, high-contrast QR code cards utilizing approximately 20% of the display widt
- When a venue Wi-Fi network is detected or configured, two stacked cards are presented:
  * 'CONNECT WI-FI': Displays the venue Wi-Fi QR code, network SSID, and password for one-scan automatic network connection.
  * 'SCAN TO SIGN UP': Displays the Web Remote / Patron Portal QR code and direct URL for submitting song requests.
- Automatic Wi-Fi Fallback: If no venue Wi-Fi is configured or detected, the display cleanly adapts to show only the single sign-up QR code card

Section: Pre-Show Screen / Singer Welcome Screen Hold & Sequenced Release

[Update Details]

To prevent audience projection distraction and keep Wi-Fi onboarding instructions visible before the show starts, singer welcome screens are hel

1. Welcome Screen Suppression:

- While the Pre-Show screen ('Hold Screen') is actively displayed on the audience projection monitor, individual 'Welcome to our new performer
- Any new performers who join the rotation (via patron smartphone QR portal, tablet kiosk, or KJ queue addition) are registered in the backgrou
- If Pre-Show Mode is engaged while a welcome screen was actively displaying, that welcome screen is immediately hidden and saved at the h

2. Sequenced Release on Pre-Show Close:

- When the host closes/turns off Pre-Show Screen mode to kick off the show, all held singer welcome screens automatically display one after a
- Once all queued new performers have been welcomed, the projection window transitions smoothly to the active Rotation Billboard queue.

Section: Dynamic Special Event Announcement Banner Studio

- Dynamic Announcement Banner: DJs can now select 'Announcement' from the Special Event Banner picker in both Lyracist and
- Host Laptop Interactive Modal: Selecting 'Announcement' immediately opens an intuitive prompt dialog where the DJ can type ar
- Remote DJ Board Support: On the web-based Remote DJ Board (/dj), tapping the 'Announcement' special event button presents
- Cancellation Safety: Dismissing or cancelling the announcement prompt cleanly preserves the previously selected banner withou

Section: Clear 'Last Round Done' Accidental Flag & Restore Performer

Lyracist Pro Suite & KSRotation: Clear 'Last Round Done' Accidental Flag & Restore Performer (`RotationViewModel.cs`, `EditSin

[Overview & Purpose]

During the final round of the evening (when 'Last Round' is engaged), performers who have completed their song are marked as c

[Interactive Badge & Direct Action]

- Clickable Badge (□): The 'DONE (LAST ROUND)' badge displayed next to a performer's name is now an interactive button badg
- Immediate Eligibility Restoration: Clearing the flag instantly restores the singer's eligibility as the Next Performer ('IsNext'), recalc

[Multiple Access Points Across Apps]

1. Lyracist Desktop:
   - Click the 'DONE (LAST ROUND) □ ' badge directly on the singer row in RotationPage or KaraokePage.
   - Right-click the singer and select '□ Clear Last Round Done Status' from the context menu.
   - Click 'Edit Performer' (□) and uncheck the 'Completed in Last Round' checkbox in the dialog.
2. KSRotation Desktop:
   - Click the 'DONE (LAST ROUND) □ ' badge on the singer row in the active rotation list.
   - Right-click the performer and choose '□ Clear Last Round Done Status' from the context menu.
3. KSRotation.Maui:
   - Tap the 'DONE (LAST ROUND) □ ' badge on the performer card.
   - Tap 'Edit Performer' (□) and uncheck 'Done in Last Round (uncheck to restore)'.
4. Remote DJ Web Portal (/dj):
   - Mobile DJs can tap the 'DONE (LAST ROUND) □ ' badge on their tablet or phone to restore performers remotely.

Section: Move to Top of Rotation & Performer Priority

Lyracist Pro Suite & KSRotation: Move to Top of Rotation (`RotationViewModel.cs`, `RotationPage.xaml`, `KaraokePage.xaml`, `M

[Overview & Purpose]

Hosts frequently encounter scenarios where a scheduled performer was temporarily away (e.g. in the restroom, stepping outside,

The new 'Move to Top of Rotation' feature instantly relocates any performer to the head of the waiting queue. If an active singer is

[Anchor Preservation Guarantee]

Crucially, moving a performer to the top of the rotation queue does NOT in any way change, reassign, or alter the Rotation Ancho

[Automatic Flag Clearing & Restoration]

If the performer being moved to the top was previously skipped ('IsSkipped'), paused ('IsPaused'), or marked inactive ('IsInactive')

[Multiple Access Points Across Apps]

1. Lyracist Desktop:
   - Singer ContextMenu: Right-click any singer row in RotationPage or KaraokePage and select 'Move to Top of Rotation'.
   - Performer Row Button: Click the dedicated 'Move to Top of Rotation' button (upload arrow icon) on any singer row.
   - Queue Toolbar: Select a performer and click the 'Move Selected Singer to Top of Rotation' button on the toolbar next to Move
2. KSRotation Desktop:
   - Performer ContextMenu: Right-click any performer row and select 'Move to Top of Rotation'.
   - Performer List Button: Click the dedicated '□ ' button on any performer row next to the Move Down button.
3. KSRotation.Maui:
   - Tap the dedicated '□ ' button on any performer card in the active rotation queue.
4. Remote DJ Web Portal (/dj):
   - Supports 'move-to-top' and 'move-top' remote control actions.

Section: New Singer Placeholder Cleanup & Automatic Row Removal

Lyracist Pro Suite & KSRotation: New Singer Placeholder Cleanup & Automatic Unused Row Removal (`NameFormatting.cs`, `Si

[Overview & Purpose]

When adding a new performer to the rotation queue (via the 'New Singer' button), a new row is created with the default placeholde

[Automatic Row Removal on Lost Focus]

If the host adds a new singer and navigates away or moves focus without typing a new name (so that 'New Singer' remains the or

[Smart Placeholder Stripping & Proper-Casing]

If the placeholder 'New Singer' is present alongside additional text (for example, typing 'New Singertom', 'New Singer Tom', or 'To

[Keyboard & Flow Handling]

Pressing Enter in the singer name input commits the name and advances focus to the next field (e.g. Song title), triggering instant

Section: Test Mode Settings GroupBox & Performer Count Options

Lyracist Pro Suite & KSRotation: Test Mode Settings GroupBox & Performer Count Options (3, 5, 10, 15) (`RotationHelpers.cs`, `M

[Overview & Purpose]

Hosts and KJs frequently utilize Test Mode to verify audio leveling, screen projection, rotation advancing, and network request kio

[Configurable Performer Counts (3, 5, 10, & 15)]

Hosts can now select exactly how many sample performers to load from a dedicated dropdown selector:

- 3 Singers: Ideal for quick audio and dual-screen verification checks.
- 5 Singers: Perfect for testing rotation handoffs and next-singer notifications.
- 10 Singers: Well-suited for testing multi-round progression and last round workflows.
- 15 Singers: Thorough full-queue testing across multi-monitor displays, billboards, and remote DJ boards.

[Dedicated Test Mode Settings GroupBox (GBX)]

- Dedicated UI Organization: Test Mode settings are now cleanly isolated within their own GroupBox ('Test Mode Settings') in both
- Instant 'Load Test Singers Now' Action: In addition to startup seeding, hosts can immediately populate or refresh the rotation que
- Preservation of Session Settings: General session settings (e.g. 'Block Duplicate Songs in Session') remain in 'Rotation Settings

[Randomized Performer Names & Song Pool]

The test generator utilizes a curated pool of 18 realistic performer names (Brenda Bumps, James Smith, Raymond Carter, Sharon

Section: Mouse Drag-and-Drop Performer Reordering

[Overview & Purpose]

Hosts and KJs frequently need to manually adjust queue order during a live show - for example, accommodating a singer stepping

[Drag Grip Handle & Visual Ergonomics]

1. Dedicated Drag Grip Handle (□ □): Each performer row in KSRotation and Lyracist features a dedicated drag grip handle on th
2. Flexible Grab Zones: In addition to the drag grip, hosts can click and drag anywhere on non-interactive portions of the performe
3. Interactive Controls Protection: Text inputs (Name, Duet Partner, Song, Artist) and interactive controls (Buttons, Checkboxes) re

[Intelligent Midpoint Drop Calculation & Boundary Protection]

1. Top/Bottom Half Precision: When dragging a performer over another row, dropping on the upper half places the performer abov
2. Empty Space Drop: Dropping a performer below the queue in the empty list space automatically relocates them to the end of th
3. Partition Boundary Enforcement: Active and inactive partitions are strictly protected. Inactive singers cannot be dragged into the
4. Linked Duet Partner Integrity: If a linked performer is reordered, EnforceLinkedAdjacency automatically keeps linked duet partn
5. Live Real-Time Propagation: Dropping a performer immediately recalculates wait times, updates next-singer blue highlights, ref

Section: New Singer Welcome Screen / Focus-Based Activation

[Overview & Purpose]

When the 'Show Welcome to our new performer screen when a new singer joins' feature is enabled in Settings, KSRotation celeb

[Focus-Based Activation Behavior]

1. Typing Interruption Elimination: The 2.5-second keystroke timer has been eliminated. The DJ can take as much time as needed
2. Natural Completion Detection: Moving focus away from the singer name text box (by pressing Tab, pressing Enter to advance t
3. Empty / Canceled Row Protection: If a new row is added and left blank or unaltered as 'New Singer', losing focus cleanly delete
4. Duplicate Protection: Once a singer is welcomed upon name entry completion, subsequent focus changes on that performer ro

Section: Filler Automix, Sample Pad & Performance Recording

[Overview & Purpose]

Lyracist v26.10.7 introduces comprehensive live performance tools:

1. Filler Automix & Loudness Normalization:
   - Per-Track Loudness Leveling: Matches the primary karaoke player standard using EBU R128 integrated loudness measureme
   - Configurable Crossfade: Adjust crossfade durations between 1 and 15 seconds (default 5s) in Settings.
   - Max Track Duration Automix: Configure an optional track play limit (e.g. 180s for 3-minute automixing) to keep filler music mov
   - Cue-Point Aware Mixing: Automatically skips leading silence on incoming tracks and triggers crossfades at MixOutMs - crossfa
   - Smart Shuffle: Automatically picks the next filler track matching the current track's tempo (within +-8% BPM tolerance, includin
2. 16-Pad Sample & Jingle Soundboard:
   - 16 low-latency pads in a 4x4 matrix accessible via the main navigation menu ('Sample Pad').
   - Drag and drop audio files directly from Windows Explorer onto any pad to assign.
   - Right-click to edit custom labels, pad colors, and volume levels or clear assignments.
   - Automatic fill-in background music ducking while any sample or jingle is playing.
   - Global F1-F12 hotkeys trigger pads instantly. Hotkeys are automatically guarded to prevent misfires while typing into text boxe
   - Settings persist automatically in Settings/samplepad.json.
3. Optional Gated Performance Recording:
   - Capture master stereo mixes directly from USB mixers (e.g. Yamaha MG10XU) or dedicated Windows audio recording inputs.
   - Per-Singer Consent Gating: Only performers with the 'Allow Performance Recording' checkbox enabled in their Singer Profile w
   - Automated Start/Stop: Automatically arms and records during active singer performances and stops on completion. Discards ta
   - Background Processing: Converts WAV recordings to high-quality MP3 via FFmpeg in the background and indexes them in the
   - Settings & Playback: Configure recording input device with live VU level meter, 5-second test clip recorder, and retention days

Section: Scaryoke Standalone Wheel / DJ's Choice & Monitor Selection

[Update Details]

Scaryoke Wheel now includes configurable DJ's Choice sliver inclusion and target monitor display selection.

1. DJ's Choice Sliver Toggle:
   - Operators can now toggle the inclusion of the two black "DJ's Choice" slivers on and off from the Settings & About tab.
   - When disabled, only Singer's Choice and custom categories populate the wheel.
   - When enabled, operators can customize the sliver width (Small, Medium, Large) independently.
2. Target Display Monitor Selection:
   - Operators can select any connected monitor to display the Scaryoke Wheel directly from the Title Bar dropdown or the Settings
   - Supports native dynamic monitor topology refresh when projectors or secondary displays are connected or disconnected.
   - Automatically positions or centers the wheel on the selected screen, and covers the chosen monitor when entering Fullscreen

Section: Scaryoke Standalone Wheel / DJ's Choice, Audience Projection & Monitor Sel

[Update Details]

Scaryoke Wheel now includes configurable DJ's Choice sliver inclusion, target monitor selection, and dedicated dual-monitor audi

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   - Automatically positions or centers the wheel on the selected screen, and covers the chosen monitor when entering Fullscreen
3. Dual-Monitor Audience Projection Window:
   - Operators can launch a borderless audience projection window on a secondary display or projector via the [□ Audience Scree
   - The primary window stays on the DJ laptop with the Spin button and control tabs, while the audience window displays a full-scr
   - Spins, decelerations, and winner/DJ's Choice reveals are synchronized frame-for-frame in real time across both displays.

Section: KSRotation & ScaryokeWheel / Scaryoke Wheel Integration & DJ Booth Spin S

[Update Details]

Scaryoke Wheel is now integrated into KSRotation with display screen precedence, DJ booth tablet controls, and consecutive dup

1. Master Console Activation & Screen Precedence:
   - Clicking [□ Scaryoke] on the Rotation toolbar launches the host wheel controller and immediately projects the wheel on the au
   - The audience wheel overlay uses top Z-index (135/136) and brings the display window forward (Topmost = true), taking prece
   - Unclicking the button cleanly shuts down the wheel across the host console, the audience screen, and all connected tablets.
2. DJ Booth Tablet Spin Station:
   - When Scaryoke is active, the Remote DJ board (dj.html) displays a dedicated booth spin station modal with a prominent [□ TA
   - Singers can step up to the DJ booth to tap and spin their challenge directly, while the live wheel animates on the big screen an
   - Includes toolbar [□ Spin Wheel] buttons on both the DJ screen and the host's laptop console.
3. Consecutive Duplicate Prevention Toggle:
   - Added a 'Prevent 2 singers in a row getting the same result' toggle across ScaryokeWheel, KSRotation, and Lyracist.
   - Prevents back-to-back duplicate challenge outcomes between consecutive performers.

Section: Rotation Management / Skipped Anchor Singer Rollover Restoration & Queue

[Update Details]

Resolved an issue where a rotation anchor singer who was skipped at the beginning of a show was erroneously bypassed upon ro

1. Round-Scoped Skip Lifecycle:
   - When a performer is skipped (including the rotation anchor), the skipped status is strictly scoped to the active round.
   - Advancing the rotation evaluates whether the sequence reaches or crosses the rotation anchor (IsRotationStart).
   - When the round completes (final singer in rotation finishes their song), all round-scoped IsSkipped flags are automatically clear
2. Immediate Turn Restoration & Queue Position Preservation:
   - Because the anchor singer's skip flag is cleared at round completion, they are restored to full active eligibility immediately when
   - In floating mode (FloatCurrentSingerToTop), the anchor singer remains at index 0 and becomes the active current singer inste
   - The on-deck indicator (IsNext) and wait-time projections accurately highlight the anchor singer as Up Next as the final performe

Section: Scaryoke Wheel / Spin History Persistence & Duplicate Prevention Across Sh

[Update Details]

Added persistent spin history tracking across all applications incorporating the Scaryoke Wheel (Scaryoke standalone wheel, KSR

1. Spin History Persistence Across Shutdowns:
   - Landed categories are automatically preserved in JSON settings when the wheel window is closed, dismissed, or shut down be
   - When the next performer is called up and the wheel is reopened, the prevent duplicate spins mechanism (AvoidRepeats) inspe
2. Automatic Spin History Reset on Rotation Save or Clear:
   - When the rotation queue is cleared (ClearRotation or ClearRotationQueue) or when end-of-night performance reports are save
   - This ensures each new rotation or show starts with a completely fresh, unbiased wheel history.

Section: Settings Page / Dedicated Karaoke Settings Tab & Display Scrolling

[Update Details]

Added a dedicated 'Karaoke Settings' tab in the main Settings page, enhanced wheel category configuration up to 11 categories w

1. Dedicated Karaoke Settings Tab:
   - Added a new top-level tab 'Karaoke Settings' in the Settings page to centralize all karaoke and Scaryoke wheel options.
   - Wheel Categories: Supports configuring up to 11 custom wheel categories (expanded from 8).
   - Category Styling & Colors: Allows DJs to set custom slice colors from the palette or reset to automatic app-assigned palette co
   - Wheel Rules & Toggles:
     * DJ's Choice Toggle: Include or exclude high-stakes black DJ's Choice slivers.
     * Singer's Choice Toggle: Include or exclude the green Singer's Choice segment.
     * Prevent Duplicate Spins: Avoid consecutive spins landing on the same category; remembers past spins across sessions and
   - Wheel Category Presets: Save, load, and delete named category presets.
   - Rotation & Queue Management: Convenient access to rotation rules (Float Current Singer to Top, Block Duplicate Songs in Se
2. Display Page Independent Column Scrolling:
   - Refactored the Display tab in Settings so that all 4 columns (Monitors & Screen Assignments, DJ & Event Banners, Audio Visu
   - Prevents truncated content on laptop screens and standard 1920x1080 displays with Windows display scaling (125%/150%).

Scaryoke Settings Tab & Audience Overlay Monitor Selection

Section: Settings Page / Scaryoke Settings Tab & Audience Overlay Monitor Selection

[Update Details]
Renamed the tab from 'Karaoke Settings' to 'Scaryoke Settings' to clarify its dedicated purpose, eliminated redundant Rotation

1. Renamed to 'Scaryoke Settings':
   - Clearly isolates all Scaryoke roulette configuration (wheel categories, colors, probabilities, rules, toggles, presets, and over
   - Eliminates redundant 'Rotation & Queue Rules' and 'Wait Times & Durations' sections to prevent confusion and conflicting

2. Audience Display & Target Overlay Monitor Selection:
   - Added Target Overlay Monitor selector under Audience Display & Overlay.
   - Allows DJs to project the Scaryoke wheel onto:
     * Rotation Billboard Screen (Default): Overlays over the primary singer rotation display.
     * Lyrics Projection Screen: Overlays onto the dedicated audience lyrics screen.
     * DJ & Special Event Banner Screen: Overlays onto the secondary banner screen.
     * Any connected display monitor (Monitor 1, Monitor 2, etc.) discovered by the system.
   - Automatically positions the audience overlay onto the selected monitor upon wheel launch and restores default placement
   - Includes a live '□ Open Scaryoke Wheel' test button to verify placement, physics, and projection.

Simplified Single DJ Name Entry & Guest DJ Default

KSRotation, Lyracist & KnockoutTrivia: Simplified Single DJ Name Entry & "Guest DJ" Default - (October 2026)

Section: Application Settings / DJ & Host Configuration

[Overview & Features]
Streamlined DJ name management across all applications by transitioning to a unified single DJ name entry, removing multi-n

1. Simplified Single DJ Name Entry (KSRotation):
   - Replaced the multi-DJ ListBox and Add/Remove buttons in KSRotation Settings with a clean, single "DJ / Host Name" entr
   - Streamlines the DJ Names settings column to a focused "DJ Name" GroupBox with immediate status display.
   - Operators can simply enter their active DJ name directly without needing to manage a secondary list or manually add and s

2. Universal "Guest DJ" Default:
   - When no DJ name is entered or the field is left blank, the application automatically defaults to "Guest DJ".
   - Applied consistently across KSRotation (WPF & MAUI), Lyracist, and KnockoutTrivia.
   - Reflected dynamically on TV billboard headers, crawler banners, rotation reports, and live window titles.

Active Special Event Banner Display on Rotation Screen

KSRotation & Lyracist: Active Special Event Banner Display on Rotation Screen - (October 2026)

Section: Display & Presentation / Special Event Banners & Overlays

[Overview & Features]
Ensured all active special event banners (Birthday, Wedding, Engagement, Anniversary, Last Song / Final Singer, Announcem

1. Unified Special Event Banner Full-Screen Overlay:
   - Previously, only Last Song was configured to display as a full-screen overlay on the audience rotation screen, causing othe
   - All active special event banners now trigger the high-visibility full-screen overlay directly on the rotation display screen acro
   - When any special event is selected or activated from the host window or remote DJ web station (dj.html), the rotation scree
   - When the special event is cleared or set to None, the overlay collapses cleanly and restores the live rotation performer list

2. Personalization & Real-Time Cache Invalidation:
   - Dynamic event banners with customized text (such as personalized Birthday celebrant names or custom Announcement m
   - Ensures newly rendered personalized banners immediately display with the latest text without showing stale cached frames

3. Standard Event Banner Asset Deployment:
   - All pre-packaged standard event banners (Birthday, Wedding, Engagement, Anniversary, Last Song, Announcement, Conn
   - If an asset is missing at runtime, the application generates standard placeholder banner files automatically.

Scaryoke Mode: Upcoming Singer Wheel Spin Ahead & Category Badges

KSRotation & Lyracist: Scaryoke Mode, Upcoming Singer Wheel Spin Ahead & Category Badges - (October 2026)

Section: Entertainment & Gamification / Scaryoke Mode & Singer Preparation

[Overview & Features]
Introduced comprehensive 'Scaryoke Mode' workflow enhancements allowing the host DJ or remote tablet operator to spin the

1. Scaryoke Mode Activation & Host Controls:
   - Dedicated 'Scaryoke Mode' toggle button on the main toolbar switches the rotation system into Scaryoke event mode.
   - The desktop Scaryoke Controller window provides an upcoming Singer Selector ComboBox and split spin buttons:
     * '□ BOOTH SPIN (QUIET)': Performs a fast, quiet wheel spin on the host laptop or tablet to assign the challenge category
     * '□ PROJECT SPIN': Shows the full-screen animated spinning wheel overlay on the audience projection display, announc

2. Singer List Challenge Badges & Actions:
   - Singer rows in the main rotation list display high-contrast category badges (e.g., □ [80s Hits]) in the wheel segment's distin
   - Singer rows feature an inline '□ Spin' / '□ Re-Spin' button when Scaryoke Mode is active.
   - The singer context menu includes dedicated actions:
     * '□ Scaryoke: Spin for Singer (Booth / Quiet)'
     * '□ Scaryoke: Spin & Project to Screen'
     * '□ Clear Scaryoke Challenge'

3. Live Audience Projection Display:
   - The audience rotation screen displays the assigned Scaryoke challenge category badge next to each performer's name in
   - When a performer takes the stage, their 'Now Singing' and 'Up Next' banners prominently highlight their assigned Scaryoke

4. Remote DJ Web Station (dj.html):
   - Queue cards on the DJ portal display the Scaryoke category badge in real time.
   - Each singer card provides a quick '□ Spin' button for convenient booth-side assignment.
   - The Scaryoke popup modal includes an upcoming singer selection dropdown and dual '□ BOOTH SPIN' vs '□ PROJECT

Scaryoke Batch Spin Ahead & Entire Rotation Assignment

KSRotation & Lyracist: Scaryoke Batch Spin Ahead & Entire Rotation Assignment - (October 2026)

Section: Entertainment & Gamification / Scaryoke Batch Spin Ahead & Entire Rotation

[Overview & Features]
Added an automated batch spinning feature allowing the host DJ or tablet manager to spin ahead for multiple upcoming singe

1. Batch Spin Ahead Options:
  - Host Controller Window (ScaryokeControlWindow):
    * Added '□ Batch Spin Ahead' controls directly below the main spin buttons.
    * Quick-action buttons: 'Next 3', 'Next 5', and 'Entire Rotation' to assign categories across the rotation in fractions of a seco
    * 'Skip singers who already have an assigned category' checkbox prevents overwriting singers who have already received
    * '□ Clear All' button to reset all Scaryoke categories across the rotation in one click.

2. Main Rotation Toolbar & Singer Context Menu:
  - Dedicated '□ Batch □ ' button on the main toolbar when Scaryoke Mode is active.
  - Left-clicking drops down batch spin commands: 'Spin Next 3 Singers', 'Spin Next 5 Singers', 'Spin Next 10 Singers', 'Spin E
  - The singer list row context menu includes a matching '□ Scaryoke: Batch Spin Ahead' submenu.

3. Mobile DJ Web Station (dj.html):
  - Added a '□ Batch Spin Ahead' card in the Scaryoke popup modal on mobile tablets and remote phones.
  - Remote DJs can trigger 'Next 3', 'Next 5', 'Entire Rotation', and '□ Clear All' remotely over Wi-Fi.

4. Smart Randomization & Consecutive Duplicate Avoidance:
  - Batch spins automatically cycle through wheel slices while adhering to consecutive duplicate avoidance rules, ensuring adj

Scaryoke Badge Clear, Monitor Selection, and Display Layout

KSRotation & Lyracist: Scaryoke Challenge Badge Auto-Clear, Dedicated Wheel Monitor Routing, and Display Layout Optimiz

1. Performer Challenge Badge Lifecycle & Auto-Clearing:
   - When a performer finishes their performance and is marked finished/done (via DJ console, remote portal, or auto-advance)
   - The singer's badge remains clean during subsequent wait times in the queue until the DJ or host spins the wheel for them a
   - Applies seamlessly across booth spins, audience project spins, batch spin aheads, and mobile DJ remote actions.

2. Dedicated Audience Wheel Monitor Routing in KSRotation:
   - Added an 'Audience Wheel Display & Monitor Routing' configuration section in the KSRotation Scaryoke Settings tab.
   - Allows DJs and hosts to select which physical screen or projection display will present the interactive Scaryoke Wheel to th
   - Supports multi-monitor setups with secondary audience screens, projectors, and video walls, with persistent device memor
   - Includes an 'Identify' button to flash screen numbers across all connected physical monitors for rapid configuration at venue

3. KSRotation Display Page Column Layout Optimizations:
   - Refined column width proportions on the Display settings tab to optimize layout density on standard 1920x1080 venue lapt
     * Screen Rotation column width narrowed from 1.0x to 0.7x to eliminate excess empty whitespace.
     * Columns 3 and 4 (hosting DJ & Special Event Banners) expanded from 2.0x combined to 2.5x combined (a 25% width inc
     * Special Event Banners Mapping header grid reorganized to vertically align the mapping title, 'Upload Banner', and 'Add E

Scaryoke Category Display on All Rotation Screens

KSRotation & Lyracist: Scaryoke Challenge Category Display Across All Rotation Screens & Themed Views

1. Venue Projection & Stage Display Integration:
   - Added color-coded Scaryoke Challenge Category badges ('□ {Category}') to all singer projection screens in both KSRotati
   - For Current Performer / On-Stage view: prominent badge with customized wheel slice palette background color and bold w
   - For Upcoming / On-Deck / Next Singers lists: every queued singer with an assigned challenge displays their category badg
   - Supported across all 11 themed projection styles:
     * Standard / Normal Rotation Panel
     * Retro Theatre Marquee Panel
     * Theatre Curtain Call Panel
     * Stadium Arena Jumbotron Panel
     * Vinyl Record Lounge Panel
     * Studio Disco Panel
     * Synthwave Neon Grid Panel
     * Outdoor Music Festival Panel
     * Las Vegas Slot Reels Panel
     * Classic Chrome Jukebox Panel
     * Sports Stadium Scoreboard Panel
   - Star Wars 3D Crawl: dynamically incorporates the performer's challenge category ('[□ {Category}]') trailing their name in th

2. DJ Management Console Rotation Views:
   - Lyracist Rotation Page: Each singer row in the rotation queue displays their custom color-coded '□ {Category}' badge adja
   - Lyracist Karaoke Page: Singer rotation queue items present the assigned Scaryoke badge with informative tooltips.
   - KSRotation Console: Singer cards and queue rows feature matching custom-styled category badges.

Scaryoke Category Display Post-Spin Requirement

KSRotation & Lyracist: Scaryoke Category Display Post-Spin Requirement - (October 2026)

Section: Entertainment & Gamification / Scaryoke Wheel Spin & Challenge Assignment

1. Real-Time Spin Roulette & Post-Spin Category Reveal:
   - Challenge categories are strictly assigned and revealed AFTER the Scaryoke Wheel has completed its spin animation and
   - While the wheel is actively spinning, the stage banner and result screens display 'SPINNING...' to preserve genuine roulett
   - Batch spin-ahead mechanisms that pre-assigned categories across upcoming singers in advance have been decommissio
   - The primary '□ Spin' toolbar button in KSRotation instantly triggers the active spin for the current or selected performer.
   - Hosts and DJs retain the '□ Clear All' button and per-singer context menu controls to reset challenges at any time.

Multi-Singer & Entire Rotation Sequential Spin Targeting

KSRotation & Lyracist: Multi-Singer & Entire Rotation Sequential Spin Targeting - (October 2026)

Section: Entertainment & Gamification / Scaryoke Multi-Singer & Entire Rotation Spinning

1. Multi-Singer & Entire Rotation Targeting:
   - Restored full capability to target multiple singers or the entire rotation at once ('Next 3', 'Next 5', 'Next 10', 'Entire Rotation').
   - Accessible via the main toolbar button ('□ Multi-Spin □ '), the singer context menu ('□ Scaryoke: Multi-Singer Spin'), the S
   - Operates sequentially: the wheel spins for each targeted singer one after another, building excitement and assigning/revea
   - Includes an in-flight '□ Stop' button on the controller and toolbar ('CancelBatchSpin') allowing hosts to halt multi-singer spi
   - Preserves duplicate avoidance rules across consecutive singers and respects the 'Skip singers who already have an assig